Communications and Networks

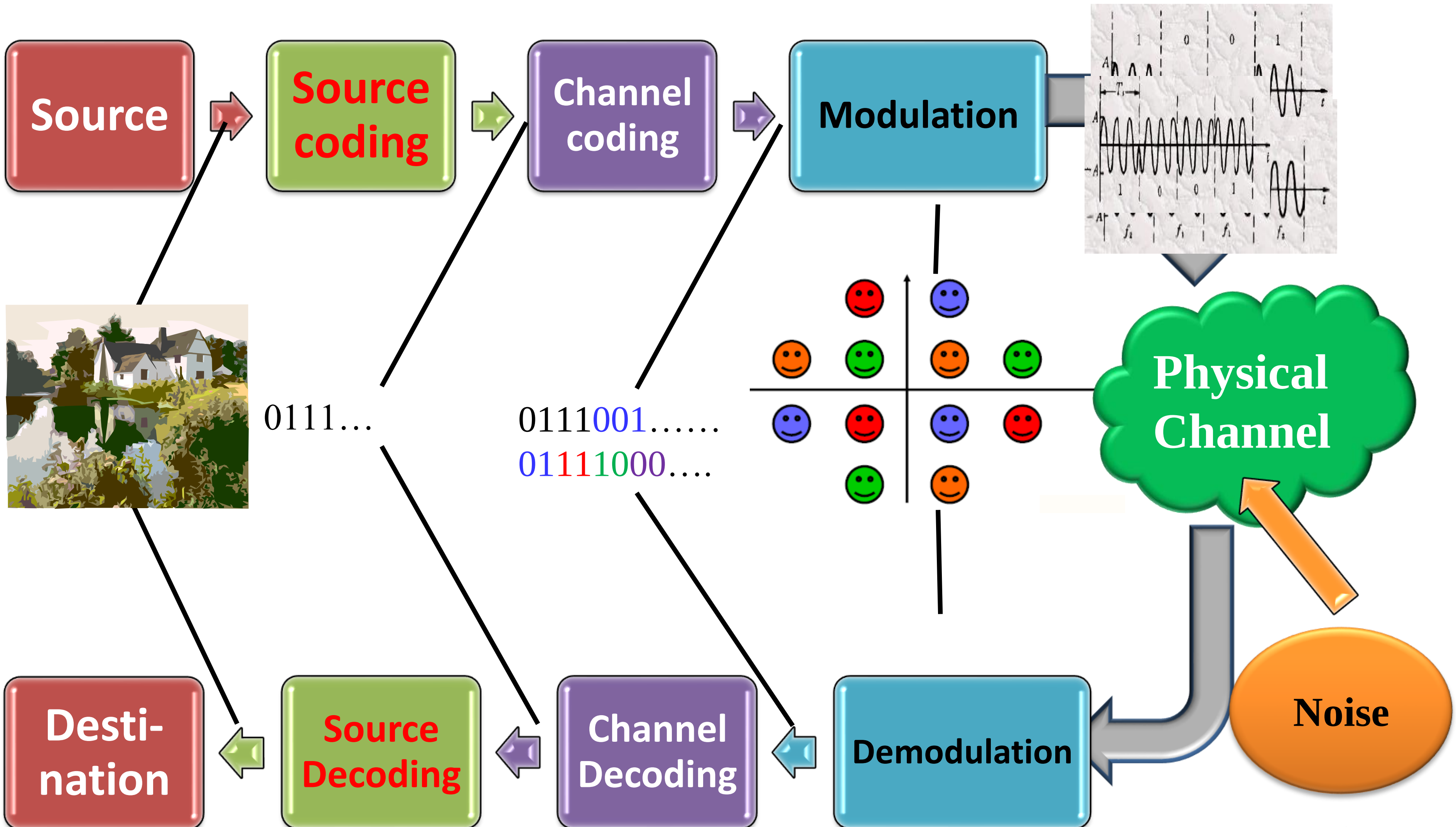
Final Overview

Sheng Zhou

About the Final Exam

- **Time and Date**
  - 06.17 Friday (9AM Beijing Time) 六教6A305
  - For those on campus, MUST attend the final offline
  - For those not on campus, let me know!
- **Q & A Session**
  - 06.10 Friday 4PM Beijing Time, FIT 3-326
  - 06.15 Wednesday 4PM, FIT 3-326

Digital Communication Link: Building Blocks



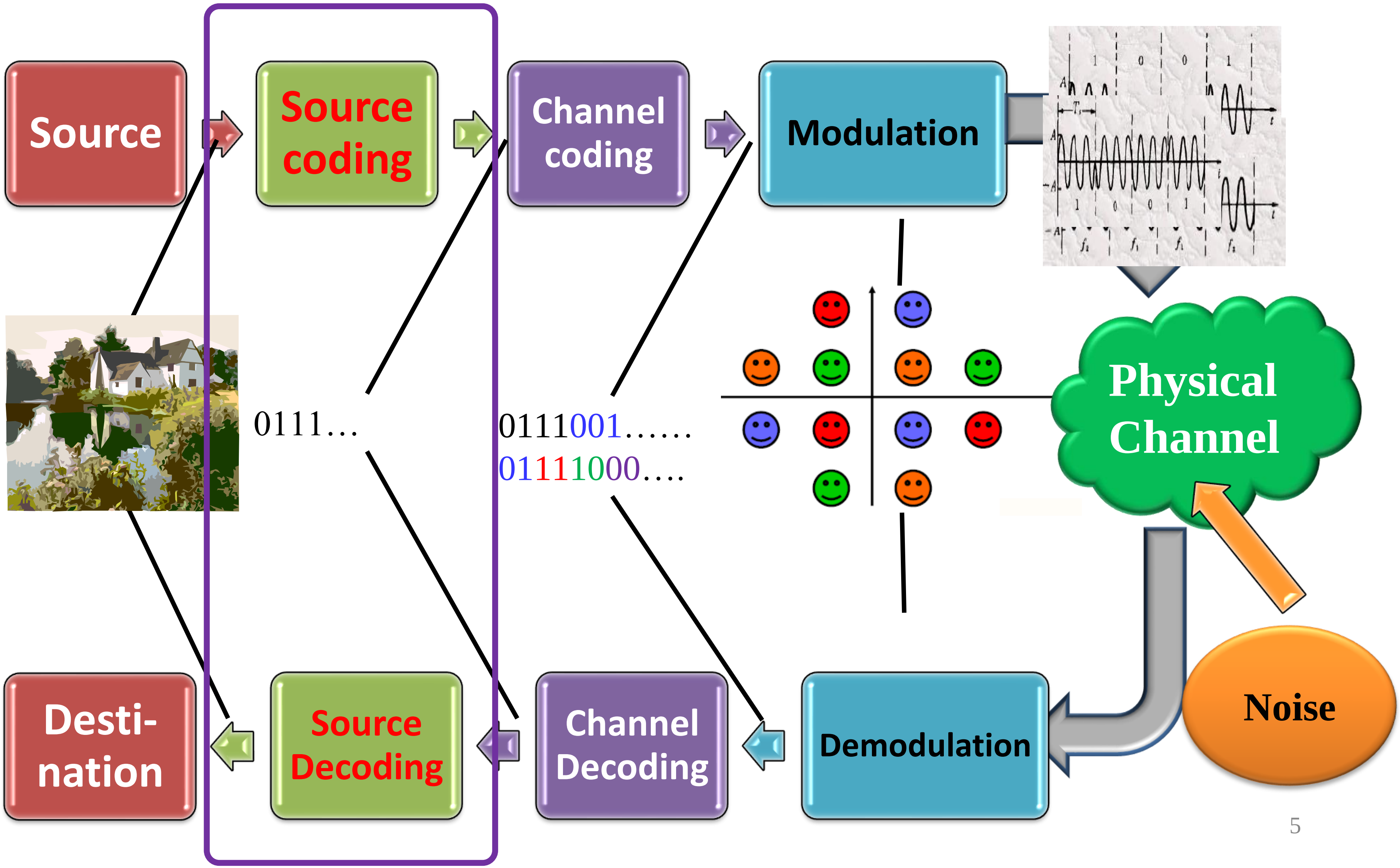
Outline

- Source Coding

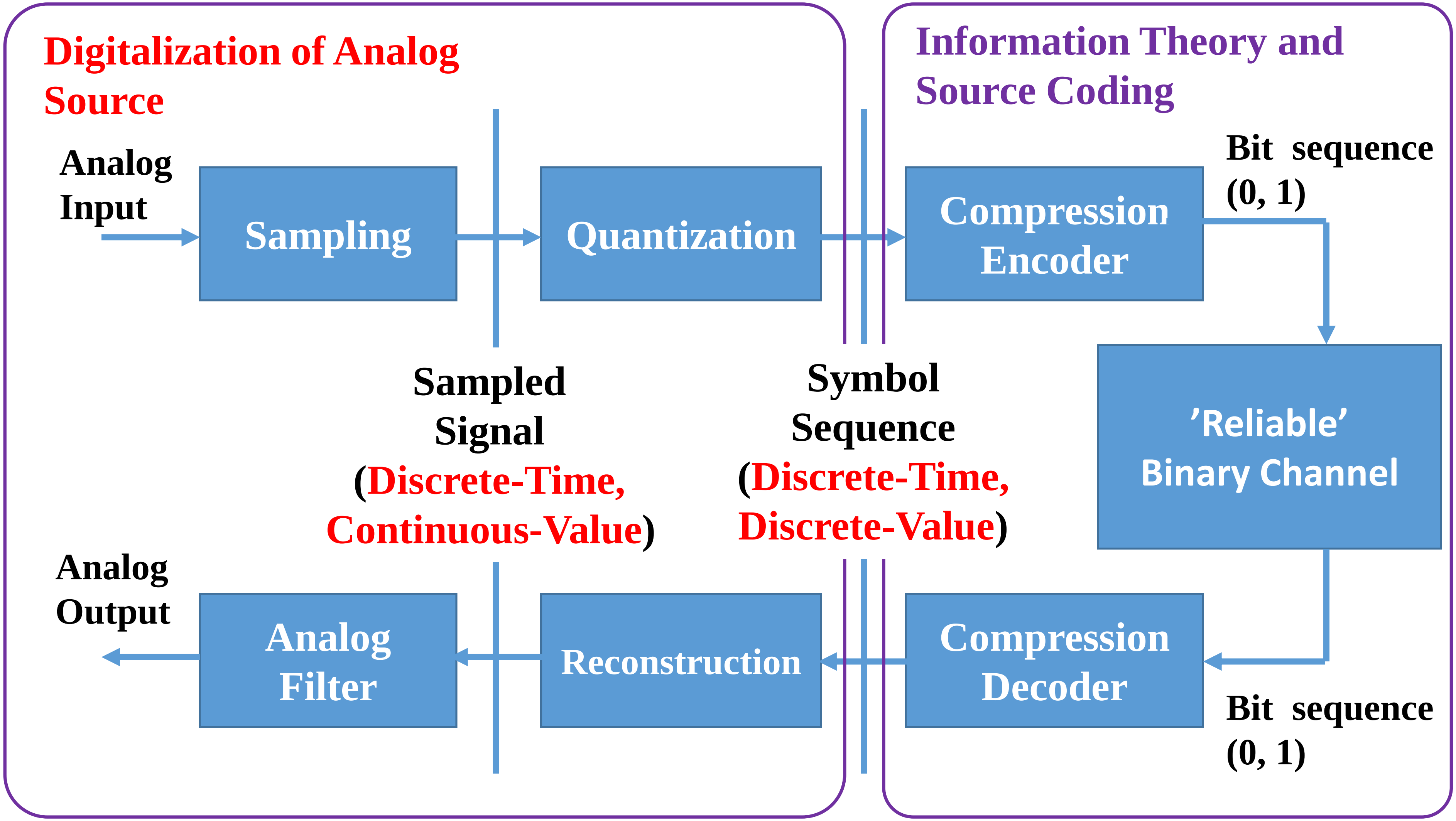
- Channel Coding

- Modulation
  - Baseband Modulation
  - Passband Modulation

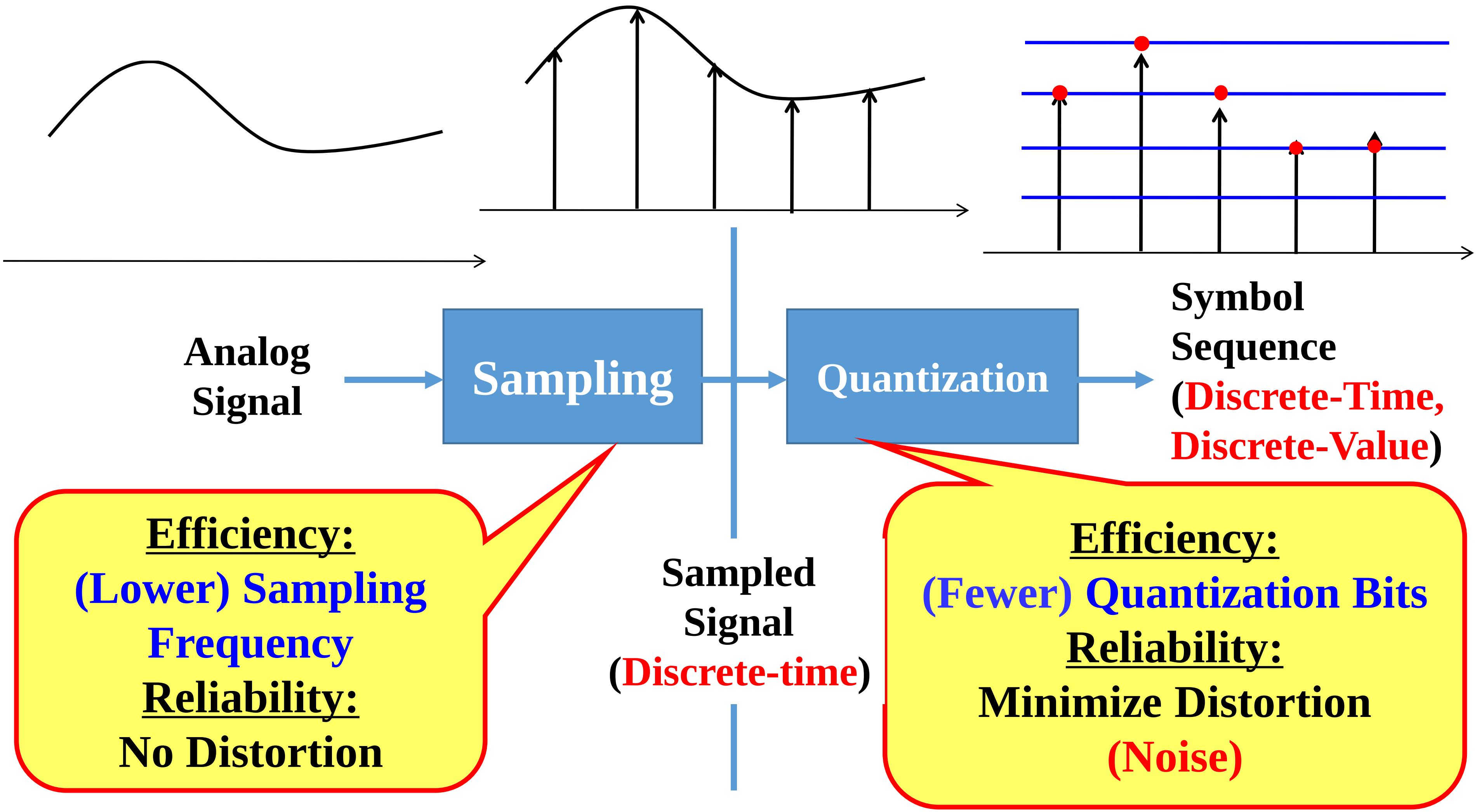
Digital Communication Link: Building Blocks



Source Coding: Overview



Digitalization of Analog Source

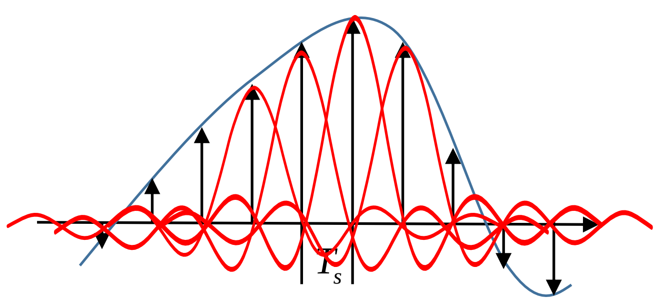


Sampling for Lowpass Signal

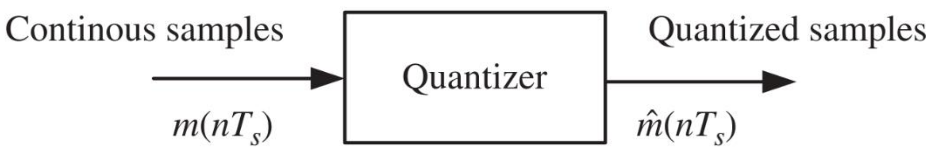
- **Sampling Theorem (Lowpass signal)**
  - A signal restricted to $[-W, W]$ is completely described by the value of the signal at periodic time instants that are separated by at most $1/(2W)$ seconds
  - Nyquist Criterion: $f_s \geq 2W$
- **Waveform Reconstruction**
  - Let $f_s = 2W$ (Nyquist rate), we have (interpolation formula)

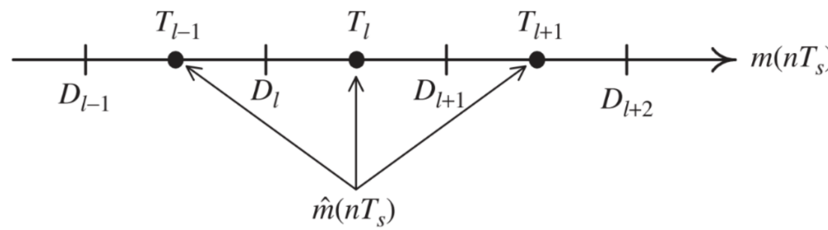
$$m(t) = \sum_{n=-\infty}^{\infty} m\left(\frac{n}{2W}\right) \frac{\sin(2\pi W t - n\pi)}{2\pi W t - n\pi}$$

$$= \sum_{n=-\infty}^{\infty} m\left(\frac{n}{2W}\right) \mathrm{sinc}(2Wt - n)$$



Quantization





- **Basic Model of Quantization**
  - Let the amplitude range of the continuous signal be partitioned into L intervals, where the l's quantization interval is determined by decision levels D_l, D_{l+1}

$$\mathcal{I}(l): \quad \{D_l < m \leq D_{l+1}\}, \;\; l \in \{1, \cdots, L\}$$

  - Step Sizes: $\Delta_l = D_{l+1} - D_l$
  - Representation Point: represents all the signal amplitudes in the interval $\mathcal{I}(l)$ by some amplitude T_l

$$\mathcal{Q}(m) = \mathcal{Q}(\{D_l < m \leq D_{l+1}\}) = T_l$$

Quantization Noise and Optimal Quantizer

- **Definitions**
  - The average quantization noise power (reliability) is given by

$$N_q = \sum_{l=1}^{L} \int_{D_l}^{D_{l+1}} (m - T_l)^2 f_m(m)\mathrm{d}m.$$

Efficiency is represented by the number of Quantization bits R: $L = 2^R$

  - Optimal quantization problem: find the set of $2L + 1$ variables $\{D_l, T_l\}$ to minimize the quantization noise power
    - The decision levels are the midpoints of adjacent representation points

$$D_l^{\text{opt}} = (T_{l-1} + T_l)/2$$

    - Each representation point is the centroid of its quantization region

$$T_l^{\text{opt}} = \frac{\int_{D_l}^{D_{l+1}} m f_m(m) dm}{\int_{D_l}^{D_{l+1}} f_m(m) dm}$$

Entropy and Information

- **Entropy of** random variable *X*: A measure of information is defined as

$$\boxed{H(X) = -\sum_{i=1}^{N} p_i \log p_i} \qquad X \sim \begin{pmatrix} x_1 & x_2 & \cdots & x_N \\ p_1 & p_2 & \cdots & p_N \end{pmatrix}$$

- Use **lossless** source coding for *X*, the average codeword length is (**efficiency**)

$$\bar{L} = \sum_{i=1}^{N} p_i l_i$$

- It is lower bounded by its entropy

$$\bar{L} \geq H(S)$$

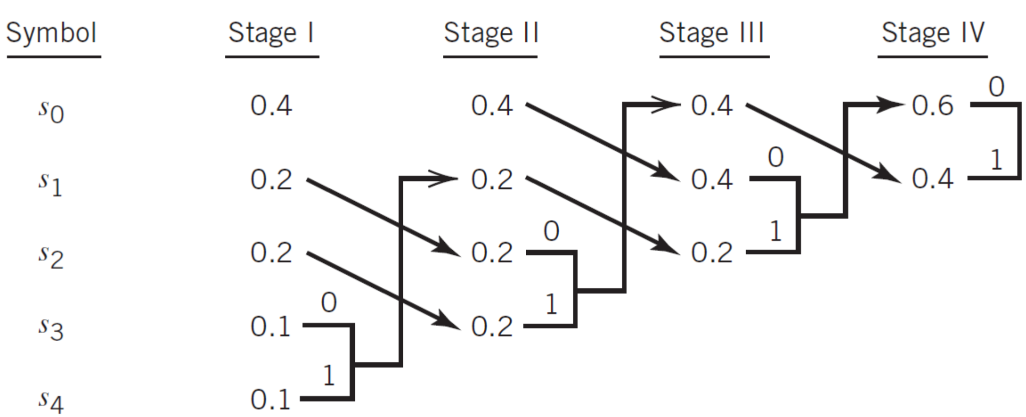
Huffman Coding

- Huffman coding is the **optimal** prefix code for discrete memoryless source, in the sense of shortest expected length.
- It also satisfies

$$H(S) \le \bar{L} < H(S) + 1$$

- An example:

$$H(S) = 2.121 \text{ bits} \qquad \bar{L} = 2.2 \text{ binary symbols}$$



Symbol	Probability	Code word
s_0	0.4	00
s_1	0.2	10
s_2	0.2	11
s_3	0.1	010
s_4	0.1	011

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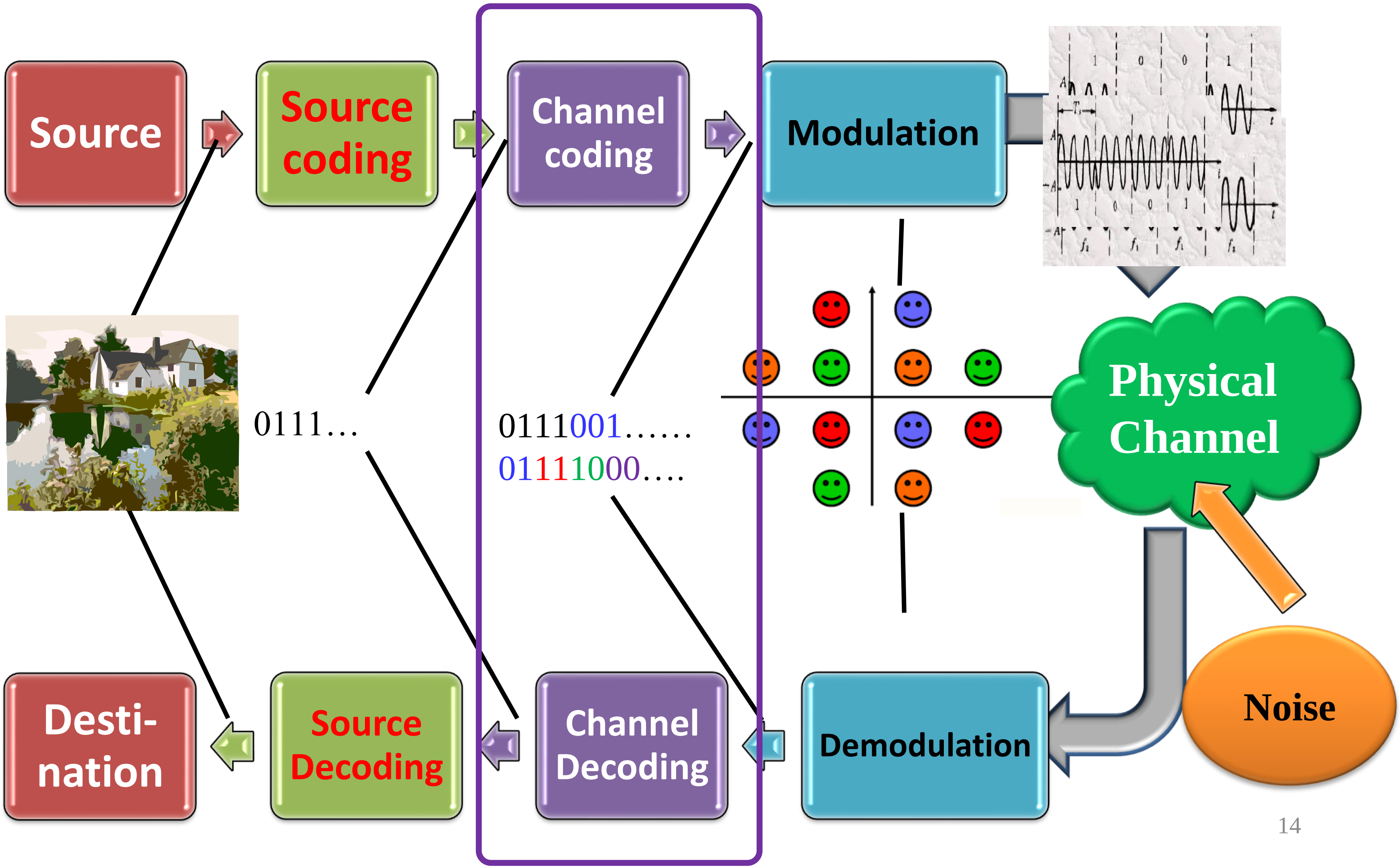
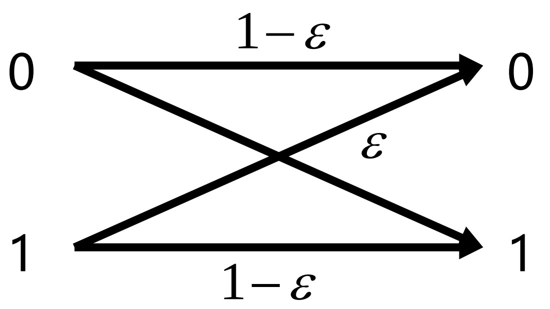


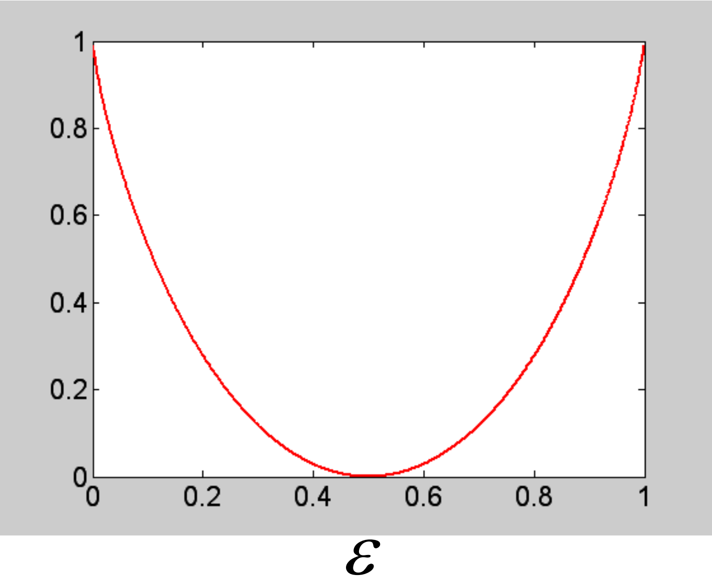
Binary Symmetric Channel

- Binary Symmetric Channel (BSC)
  - A typical model for communication channel
  - Applied in analyzing the channel coding
  - Basic model：

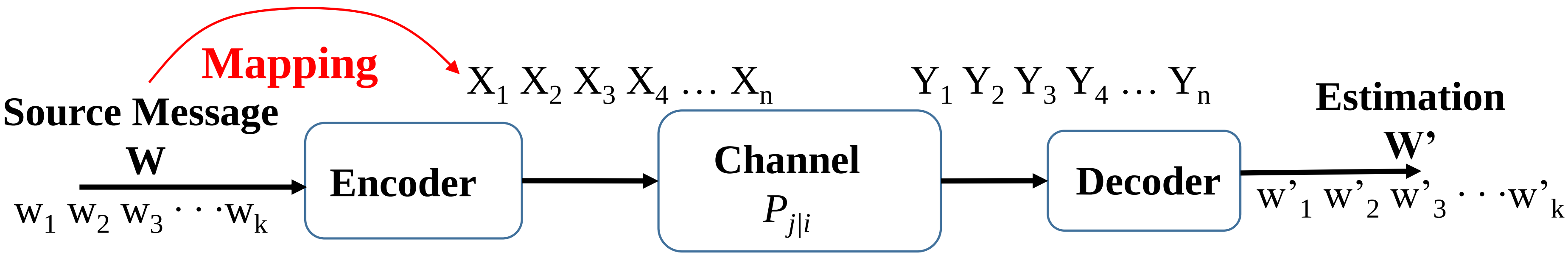


$$C = \lim_{K \to \infty} \frac{1}{K} \log_2 \mathcal{N} = \max_{p(x)} I(X;Y)$$

$$\Rightarrow \begin{cases} p = \frac{1}{2} \\ C = 1 + \varepsilon \log \varepsilon + \bar{\varepsilon} \log \bar{\varepsilon} \end{cases}$$



Channel Coding: Definition



represented by a k-bit sequence

represented by a n-bit sequence (n>k)

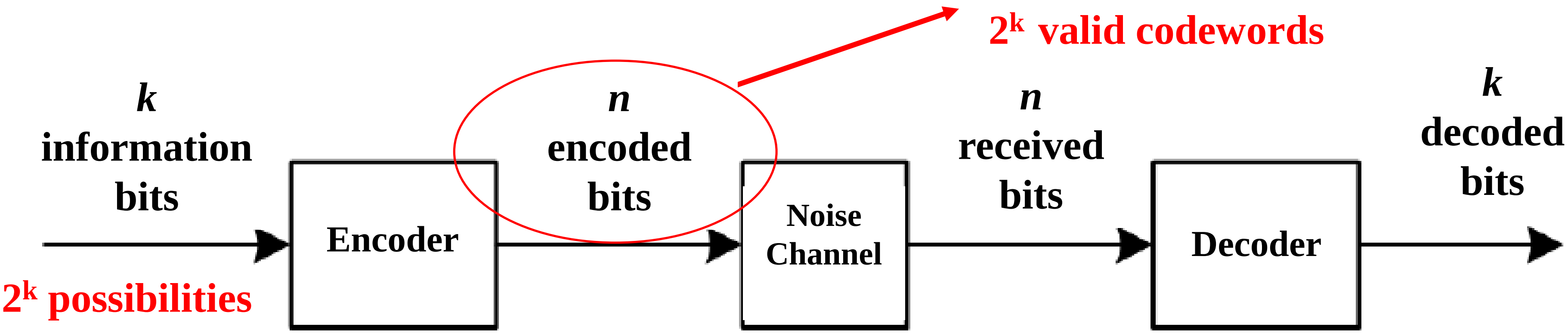
n-bit sequence received

k-bit sequence after decoding

- **Channel Coding:**
  - By properly adding redundancies, channel coding corrects possible errors in channel transmission
  - Also known as, **Error Correction Coding**

Block Code

- (n, k) block code



- The minimum Hamming distance $d_{\min}$
  - Error detection capability: e (in bits),
  - Error correction capability t (in bits)

$$\boxed{d_{\min} \geq t + e + 1}$$

$$\boxed{t \leq e}$$

Linear Block Code

- Design of linear block code
  - Encoding: Multiply the **generation matrix** **G**

$$\boxed{\mathbf{a} = \mathbf{x}\mathbf{G} = \mathbf{x}[\mathbf{I} \vdots \mathbf{Q}]}$$

***n*-dimension codeword**

***k*-dimension information vector**

G is usually full-rank. If G is in the form [I:Q], then it is a systematic code

Note: calculations are based on GF(2)

- The decision rule of error detection is whether

$$\boxed{\mathbf{b}\mathbf{H}^T \neq \mathbf{0}}$$

Identifies 2^r-1 unique error patterns

- $\mathbf{H} = [\mathbf{Q}^T \vdots \mathbf{I}_r]$ is the **typical parity check** matrix

Hamming Code

- Hamming code: Linear block code satisfying

$$2^r = n+1$$

- Below we design a Hamming code:
  - Let r=3, the codeword length $\boxed{n = 2^3 - 1 = 7}$
  - Length of information bits： $k=n\text{-}r=4$
  - Determine the dimensions of the transposed parity check matrix

$$\boxed{\mathbf{H}^T = \begin{bmatrix} & \mathbf{Q} & \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$$

Example Design of Hamming Code

- Design a (7,4) Hamming code
- **Determine Q**
  - Select rows different from 000,100,010,001
  - If we choose:
    - 101 for 1st bit error
    - 111 for 2nd bit error
    - 110 for 3rd bit error
    - 011 for 4th bit error

$$\mathbf{G} = [\mathbf{I}_k \quad \mathbf{Q}] = \left[\begin{array}{cccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{array}\right]$$

$$\boxed{\mathbf{H}^T = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$$

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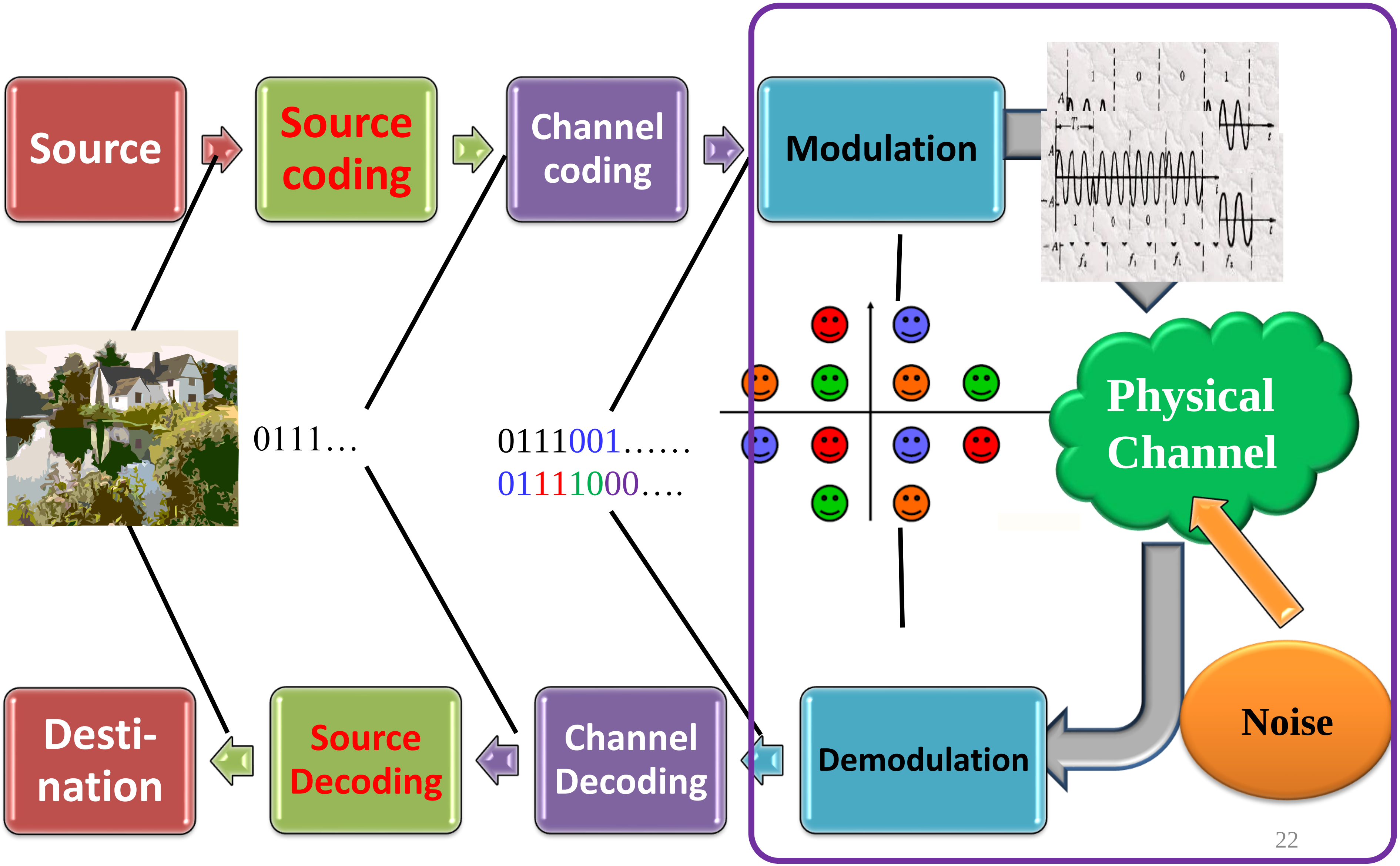
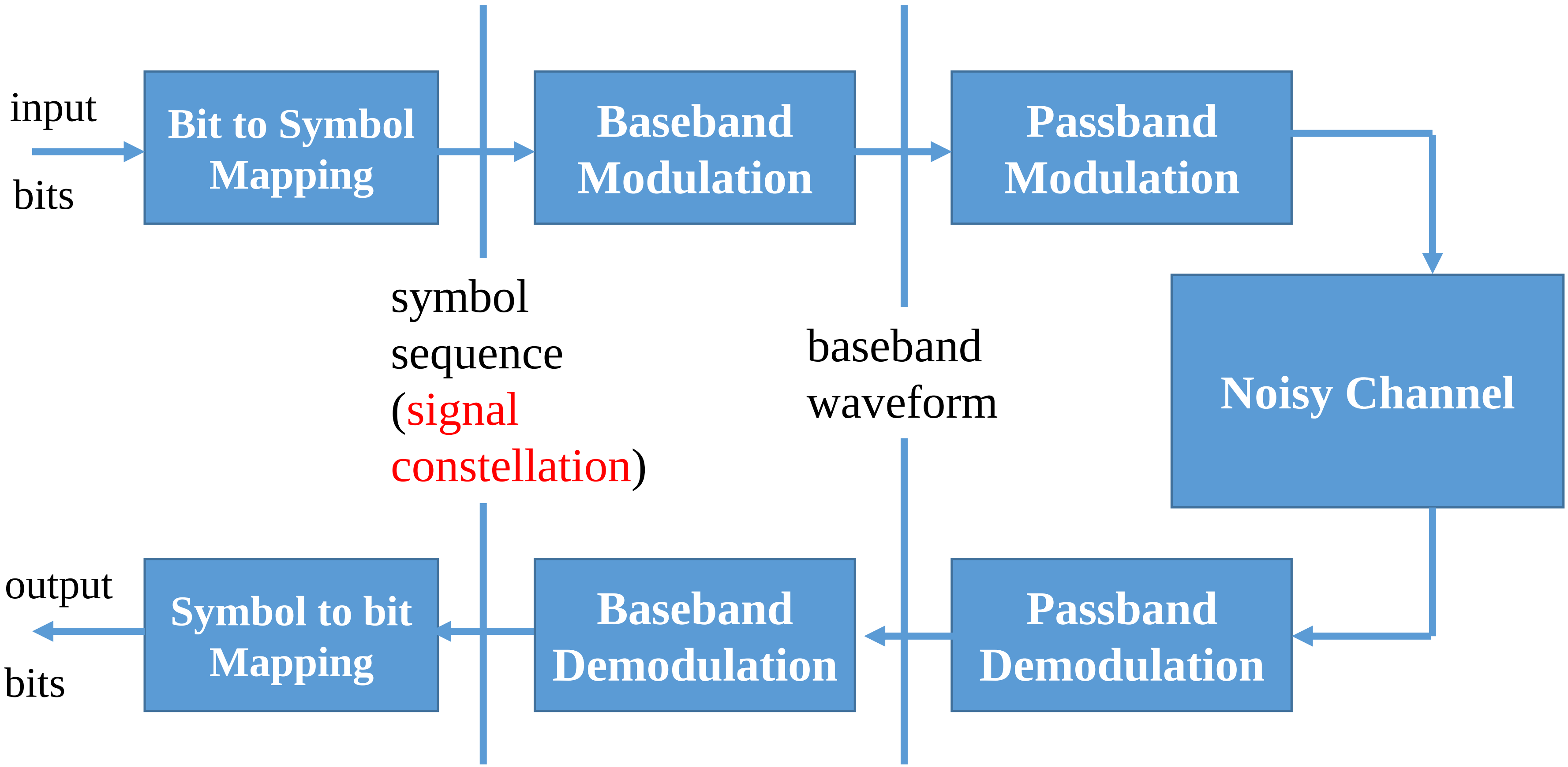


Modulation: Overview



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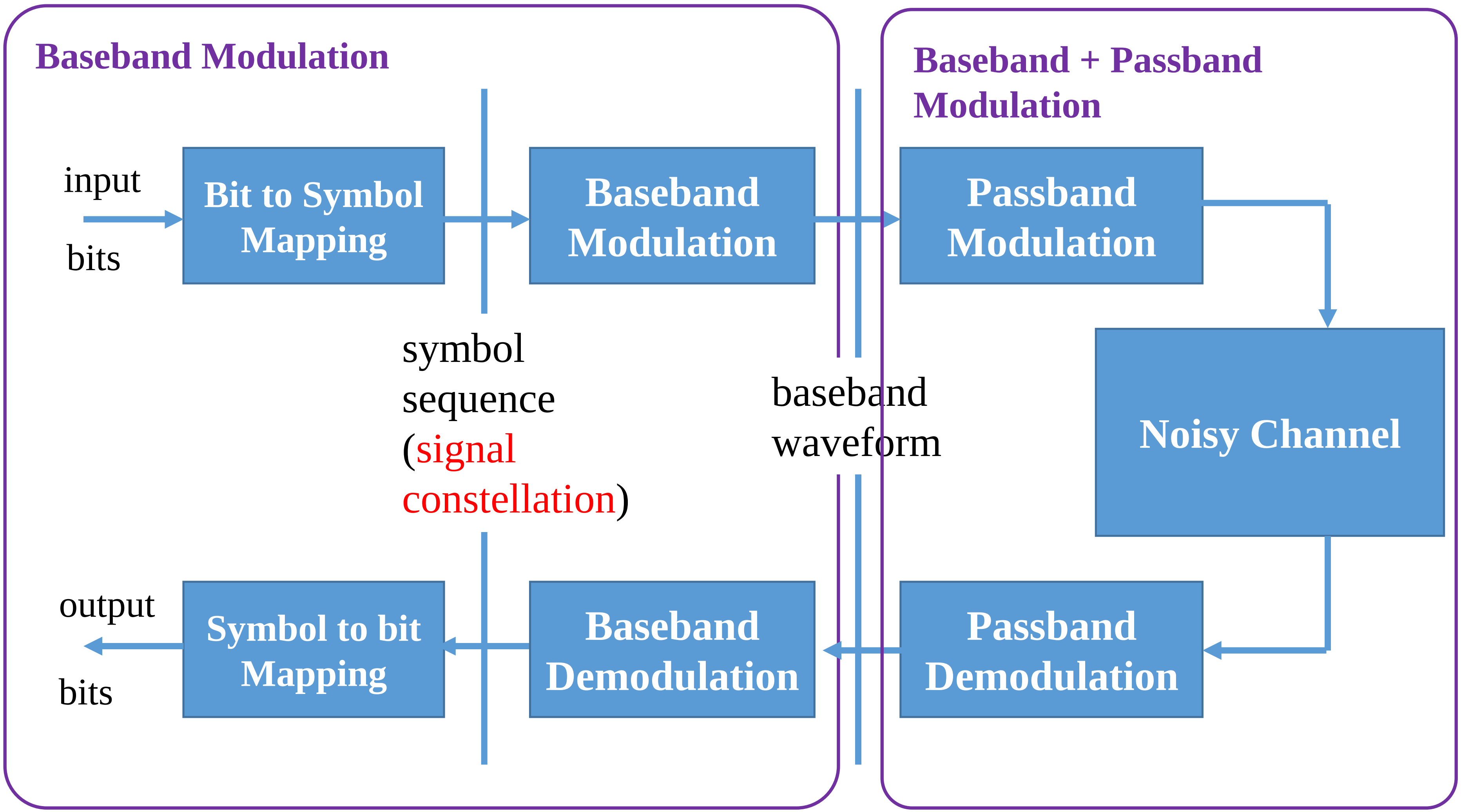

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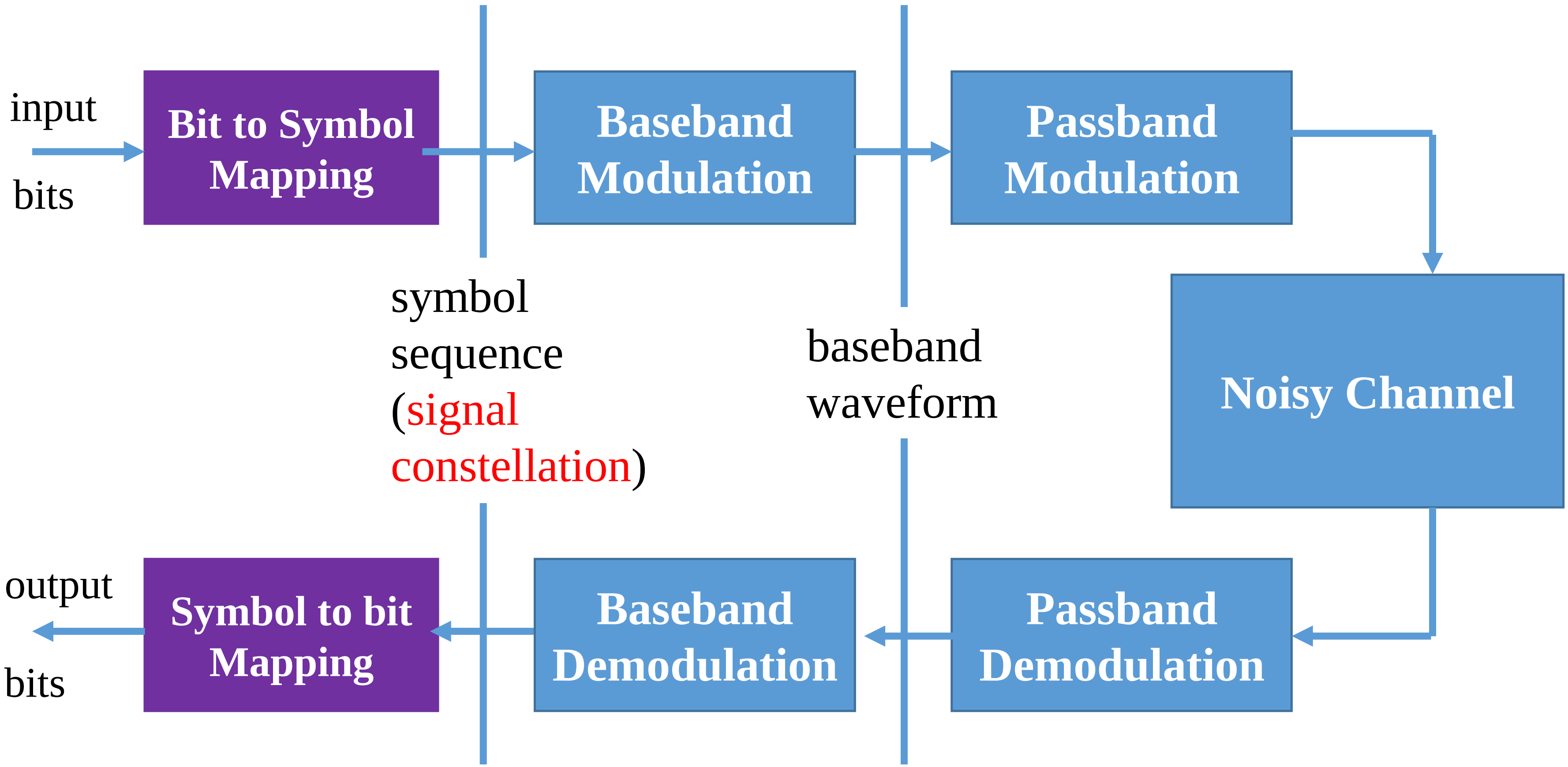


Bit Sequence to Symbols

… 011000111110101111110110010101100…

$a_1 \quad a_2 \quad a_3 \quad a_4 \quad a_5 \quad a_6 \quad a_7 \quad a_8$

- Packing r bits into one symbol
- Symbol a_k is in a finite set

$$a_k \in \mathcal{A}$$

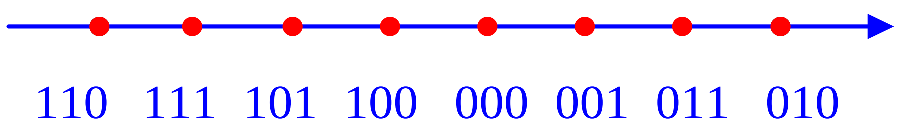
$$M =: |\mathcal{A}| = 2^r \quad \Rightarrow \quad r = \log_2 |\mathcal{A}|$$

M: number of possible symbols in the symbol set (alphabet)

Forms of Transmission Symbols (1)

- They can be real numbers

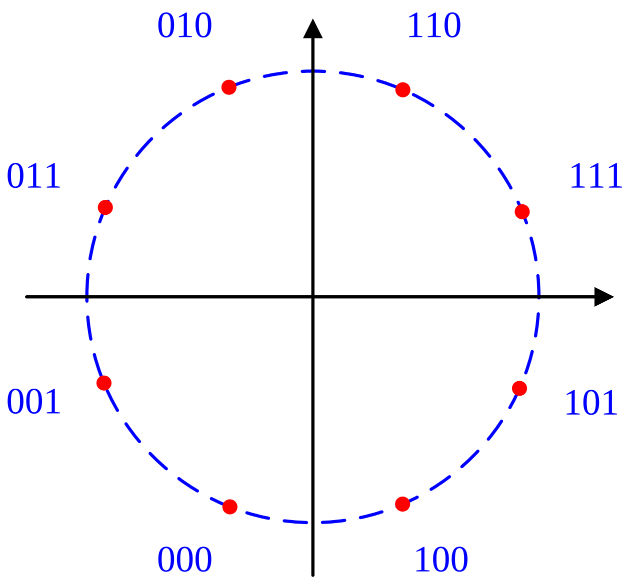
***Amplitude* Shift Keying (ASK)**

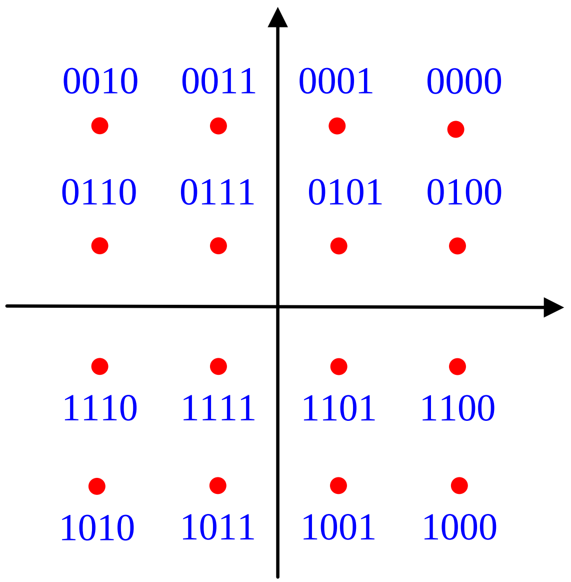


Pulse *Amplitude* Modulation（PAM）
with M = 8, r = 3 bits

Forms of Transmission Symbols (2)

- They can be complex numbers
  - Constellation





Phase Shift Keying (PSK)

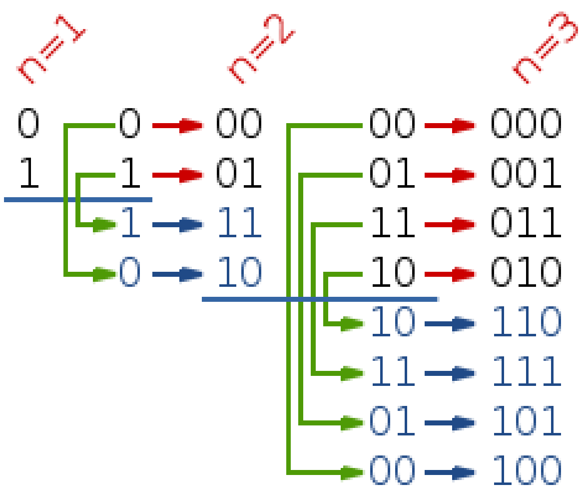
Quadrature Amplitude Modulation (QAM)

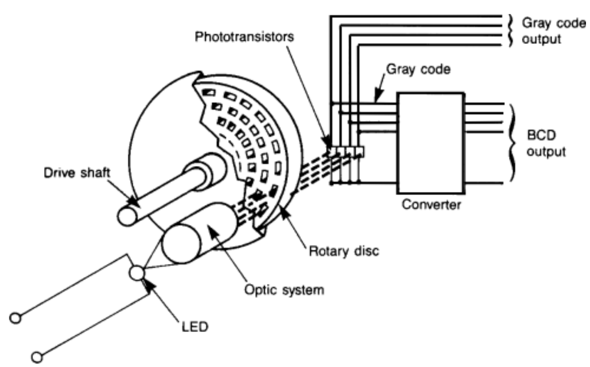
Gray Encoding

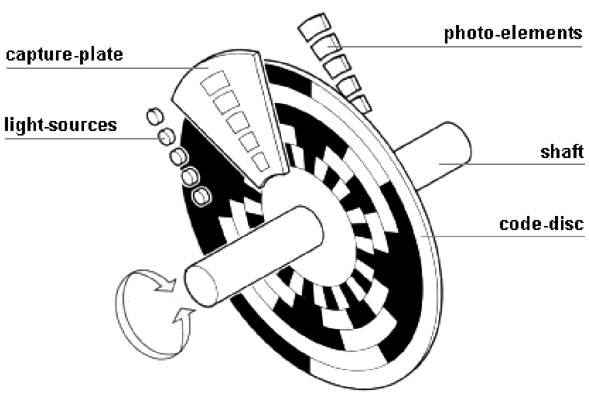
- The mapping of the r-bits to the M possible symbols, i.e.,

$$\boxed{\mathcal{G} : \{0,1\}^{\log_2 |\mathcal{A}|} \mapsto \mathcal{A}}$$

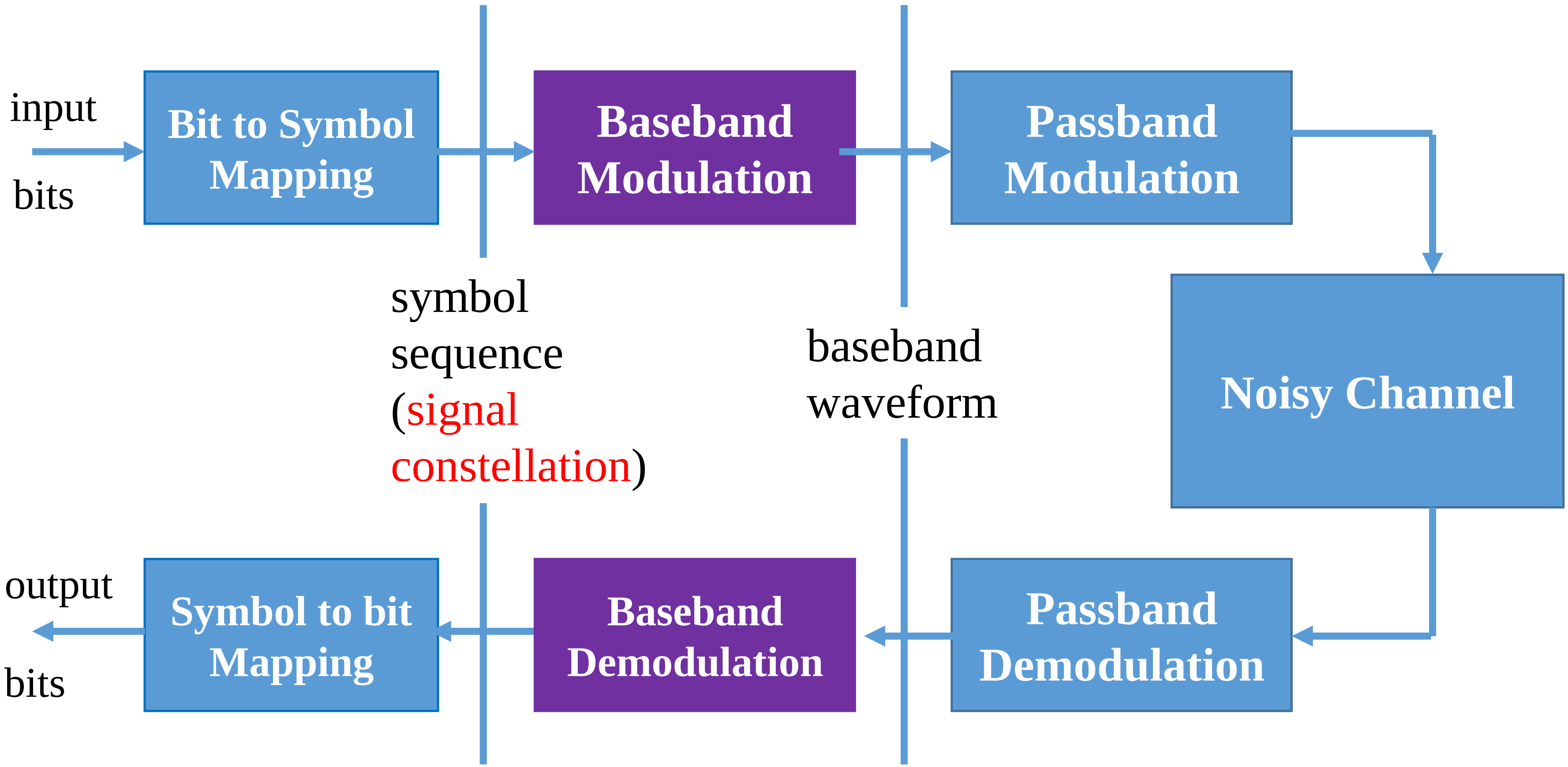
  can be done in a number of ways, in fact arbitrary
- One good choice: Gray encoding
  - **Adjacent symbols only differ by ONE bit** (WHY good?)







Modulation: Overview



Baseband Modulation: Overview

- Use the same pulse waveform for all symbols
  - With time shifts, i.e., send a symbol every T_s seconds

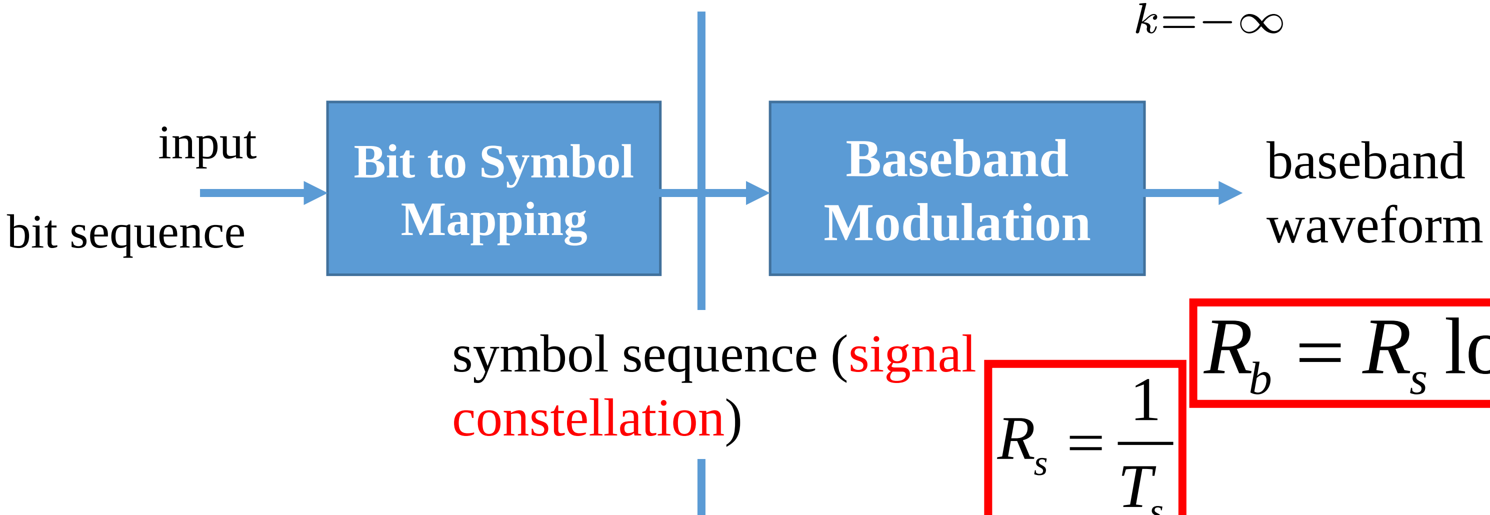
$$\phi_k(t) = g(t - kT_s)$$

- Our goal: decide T_s and $g(t)$

0110001…

$a_1, a_2, a_3, \ldots$

$$s(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT_s)$$

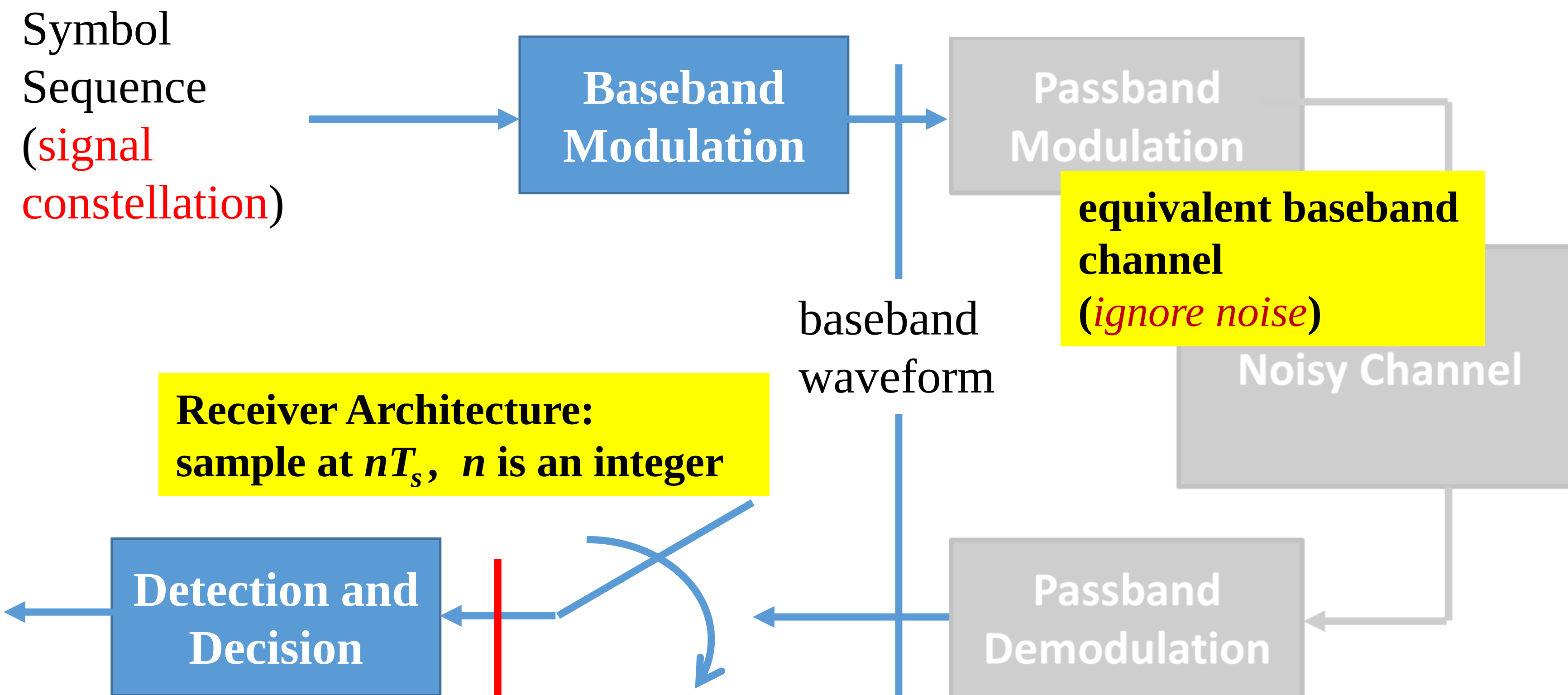


$$\boxed{R_s = \frac{1}{T_s}}$$

$$\boxed{R_b = R_s \log_2 M}$$

Receiver Architecture (direct sampling)

$$s(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT_s)$$



$$s(nT_s) = \sum_{k=-\infty}^{\infty} a_k g(nT_s - kT_s)$$

$$s(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT_s)$$

Inter-symbol Inference

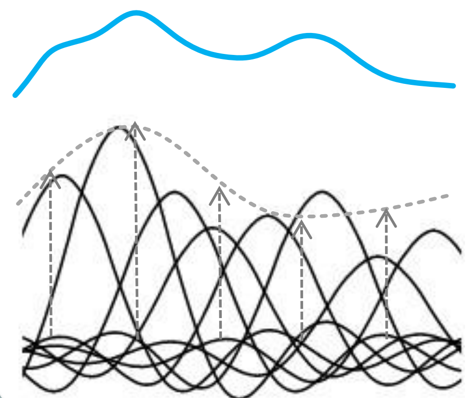
- Receiver samples at times nT_s

$$\boxed{s(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT_s)}$$

Sample at nT_s

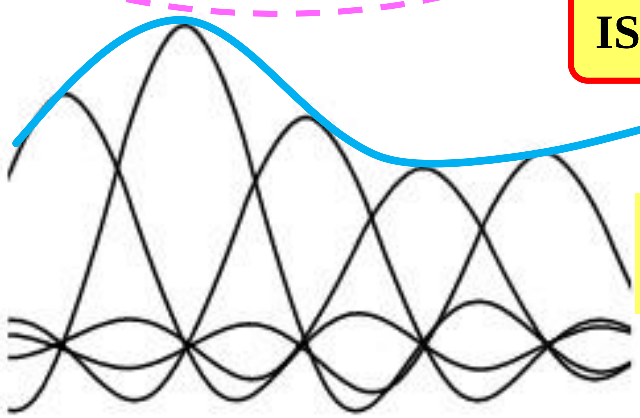
$$\boxed{s(nT_s) = a_n g(0) + \sum_{k=-\infty, k\neq n}^{\infty} a_k g\big((n-k)T_s\big)}$$

Inter-Symbol Interference (ISI)



$$\boxed{s(nT_s) = a_n}$$

ISI should be 0



Condition for 0 ISI?

Nyquist Criterion

- Look at the spectrum

$$\boxed{g(t)\sum_{n=-\infty}^{\infty}\delta(t-nT_s)=\delta(t)}$$

$$\Updownarrow\ \mathcal{F}$$

$$\boxed{G(f)*\sum_{n=-\infty}^{\infty}\frac{1}{T_s}\delta\left(f-\frac{n}{T_s}\right)=1}$$

$$\boxed{\sum_{n=-\infty}^{\infty}G\left(f-\frac{n}{T_s}\right)=T_s}$$

Shift the replicas of the spectrum of *g*(*t*) by n/T_s , where $n \in \mathbb{Z}$
The resultant spectrum is a constant, if and only if no-ISI

Data Rate & Bandwidth Efficiency

- The symbol rate must satisfy

$$\boxed{R_s \le 2W}$$

Equality Condition ?

- We also have bit $\boxed{R_b = R_s \log_2 M}$
- Therefore $\boxed{R_b \le 2W \log_2 M}$

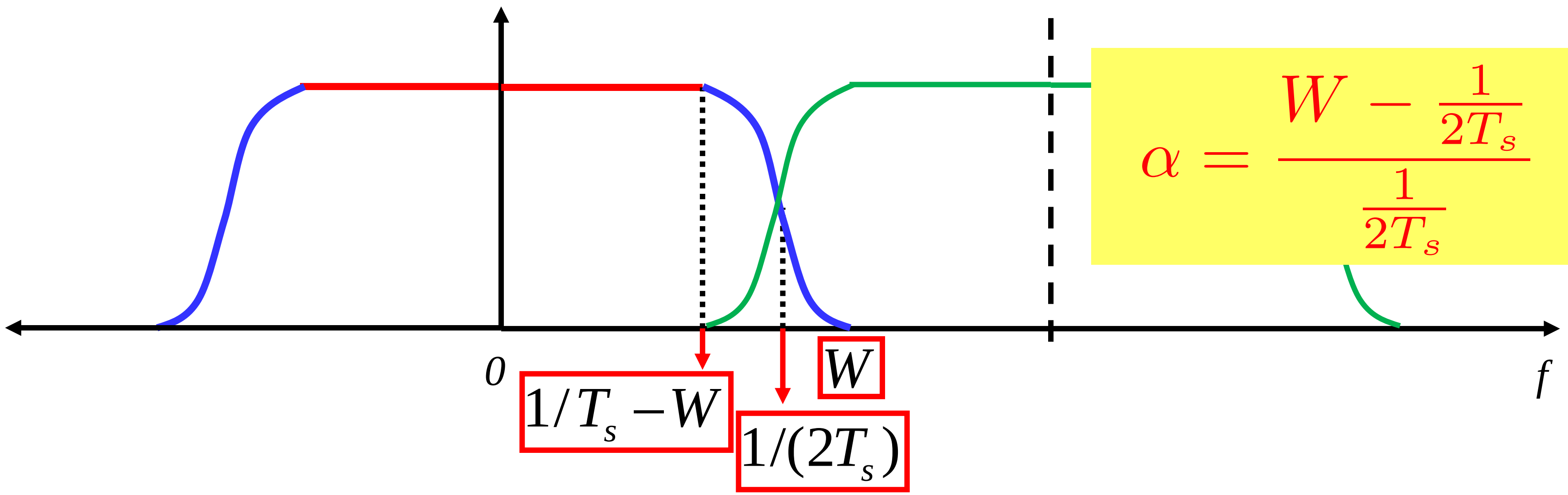
"Broadband Access"!

- Bandwidth efficiency: data rate per bandwidth

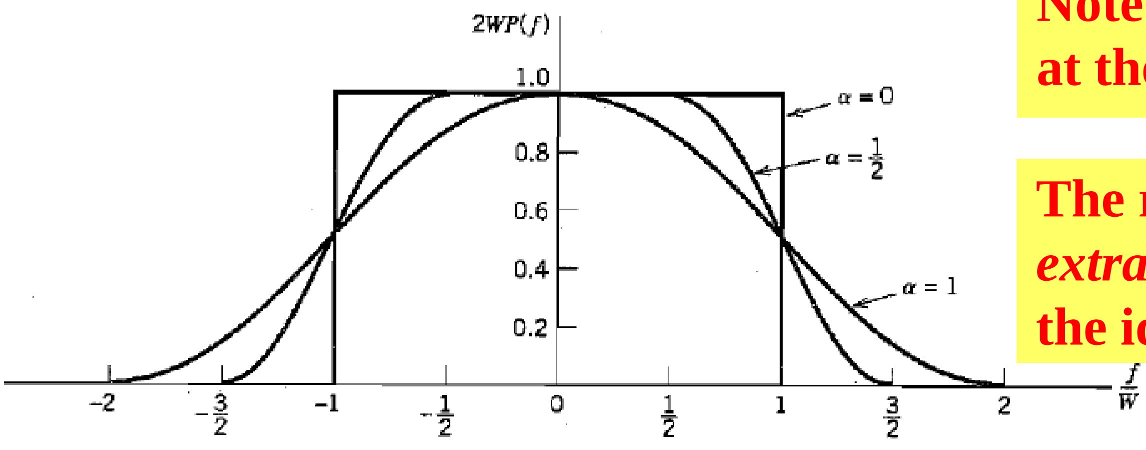
$$\boxed{\eta = \frac{R_b}{W} \le 2\log_2 M}$$

"Broadband Access"!

Raised Cosine Filter: Rolloff Factor



$$\alpha = \frac{W - \frac{1}{2T_s}}{\frac{1}{2T_s}}$$



(a)

Note that the ideal low-pass filter at the Nyquist rate is $1/(2T_s)$

The rolling factor is the unified *extra* bandwidth as compared to the ideal low-pass filter

Raised Cosine Filter: Properties

- Some simple properties
  - between

$$\boxed{\alpha = 2WT_s - 1}$$

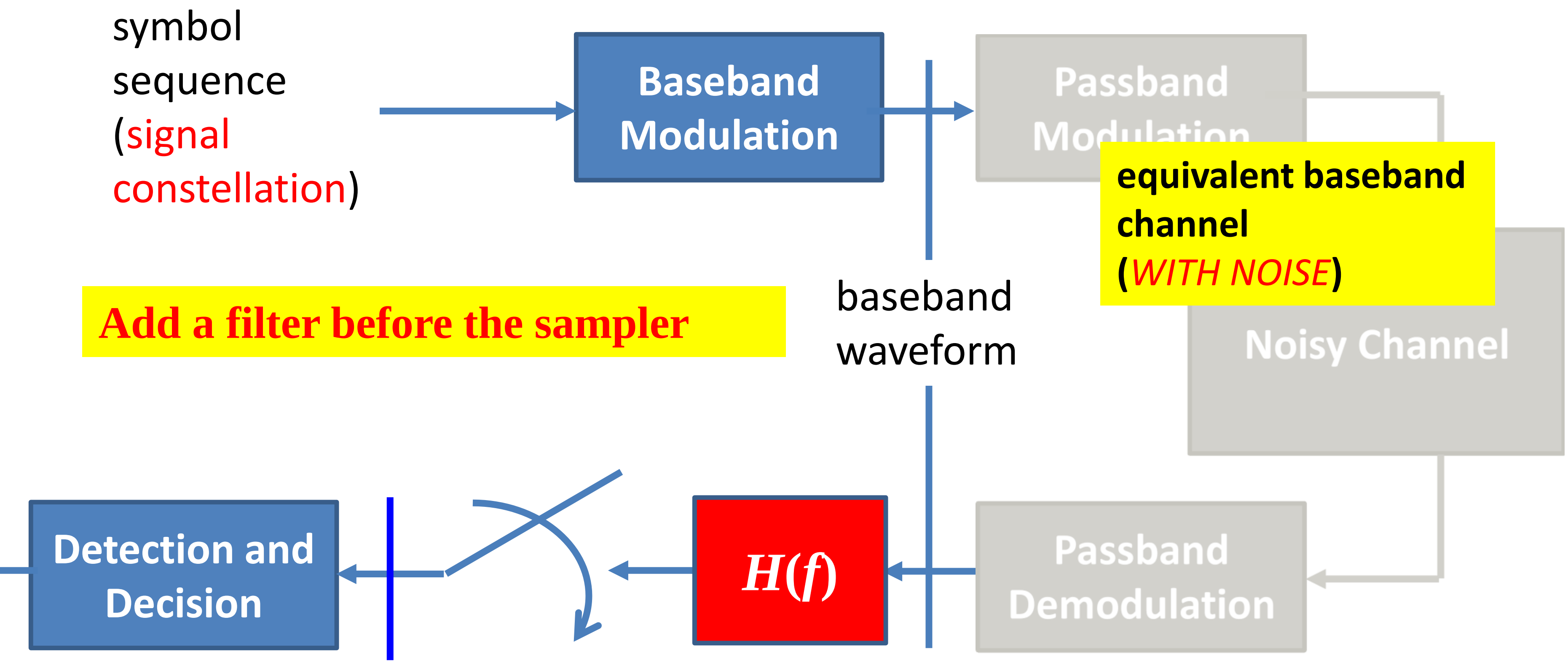
$$\Rightarrow \boxed{W = \frac{\alpha+1}{2T_s} = \frac{\alpha+1}{2}R_s} \Rightarrow \boxed{\frac{R_s}{2} \le W \le R_s}$$

$$\Rightarrow \boxed{R_s = \frac{1}{T_s} = \frac{2}{\alpha+1}W} \Rightarrow \boxed{W \le R_s \le 2W}$$

Simple ones, but can easily get confused, stick to your intuition!

Optimal Receiver Architecture (with Noise, reliability issue!)

$$s(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT_s)$$



$$\sum_{k=-\infty}^{\infty} a_k g(t - kT_s) + n(t)$$

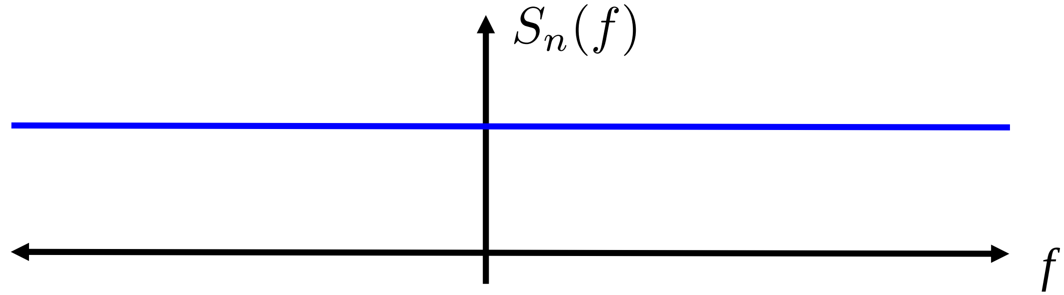
White Gaussian Noise

- We general consider *white* Gaussian noise $n(t)$
- The autocorrelation function of $n(t)$ is

$$R_n(\tau) = \frac{n_0}{2}\delta(\tau)$$

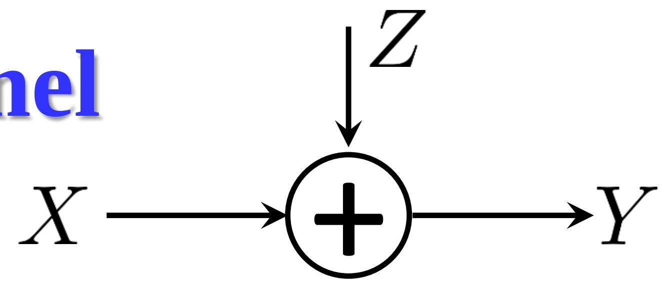
- Then the power spectrum density (PSD) of $n(t)$ is

$$\boxed{S_n(f) = \frac{n_0}{2}}$$



- PSD has all the frequency components: White!
- What is the average power of white Gaussian Noise?

Additive Gaussian Channel



- Gaussian noise Z
- Power constraint: $\mathbb{E}\{X^2\} \leq P$
- $\max\limits_{p(x)} I(X;Y) = \max\limits_{p(x)} h(Y) - \boxed{h(Z)}$

  Constant, $\frac{1}{2}\log\left(2\pi e\sigma^2\right)$

- $\mathbb{E}\left(X^2\right) \leq P \Leftrightarrow \mathbb{E}\left(Y^2\right) \leq P + \sigma^2$

$$\Rightarrow h(Y) \leq \frac{1}{2}\log\left(2\pi e\left(P+\sigma^2\right)\right) \text{ and " = " holds iif}$$

$$Y \sim \mathcal{N}\left(0, P+\sigma^2\right) \Leftrightarrow X \sim \mathcal{N}(0, P)$$

$$C = \frac{1}{2}\log\left(1 + \boxed{\frac{P}{\sigma^2}}\right)$$

γ : Signal-to-Noise Ratio (SNR)

Matched Filter

$$\mathrm{SNR} = \frac{\left| a_i \int_{-\infty}^{\infty} G(f) H(f) e^{j2\pi f T_s} df \right|^2}{\frac{n_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 df}$$

$$H(f) = bG^*(f) e^{-j2\pi f T_s}$$

$$h(t) = g(T_s - t)$$

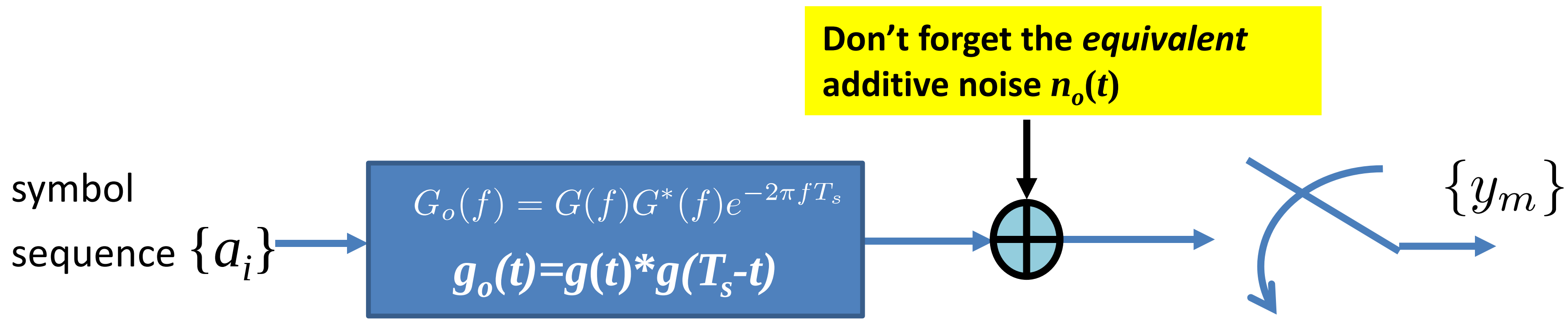
- Maximizes the SNR at the sampling output:

$$\mathrm{SNR}_{\max} = \frac{2|a_i|^2}{n_0} \underbrace{\int_{-\infty}^{\infty} |G(f)|^2 df}_{\int_{-\infty}^{\infty} |g(t)|^2 dt}$$

Average SNR

$$\gamma = \frac{2E_s}{n_0}$$

Nyquist Criterion: Revisited



- Recall the Nyquist Criterion is described in the frequency domain
- The following frequency response

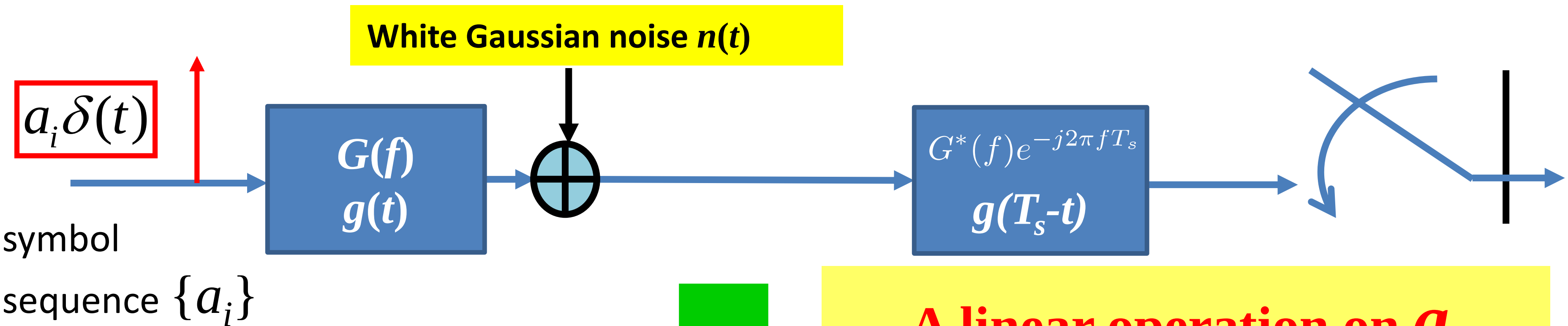
$$G_o(f) = |G(f)|^2 \, e^{-j2\pi fT_s}$$

should satisfy the Nyquist Criterion

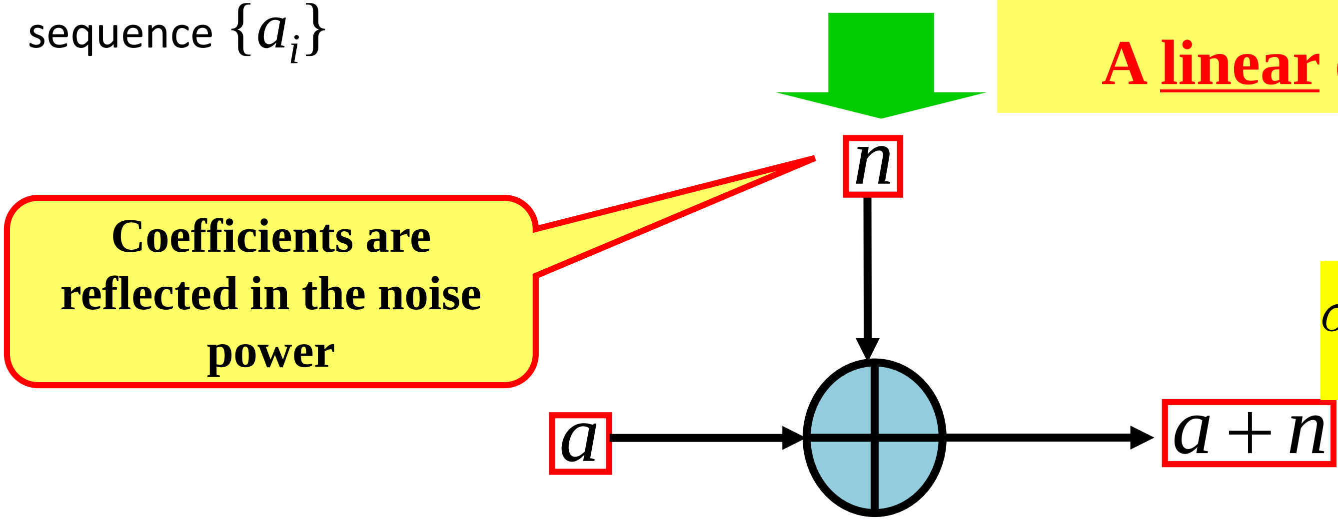
Therefore, *G*(*f*) should satisfy the <u>squared root</u> Nyquist Criterion

Equivalent Baseband Transmission Model

- Through squared root Nyquist waveform filters: No-ISI, we can again focus on each symbol separately



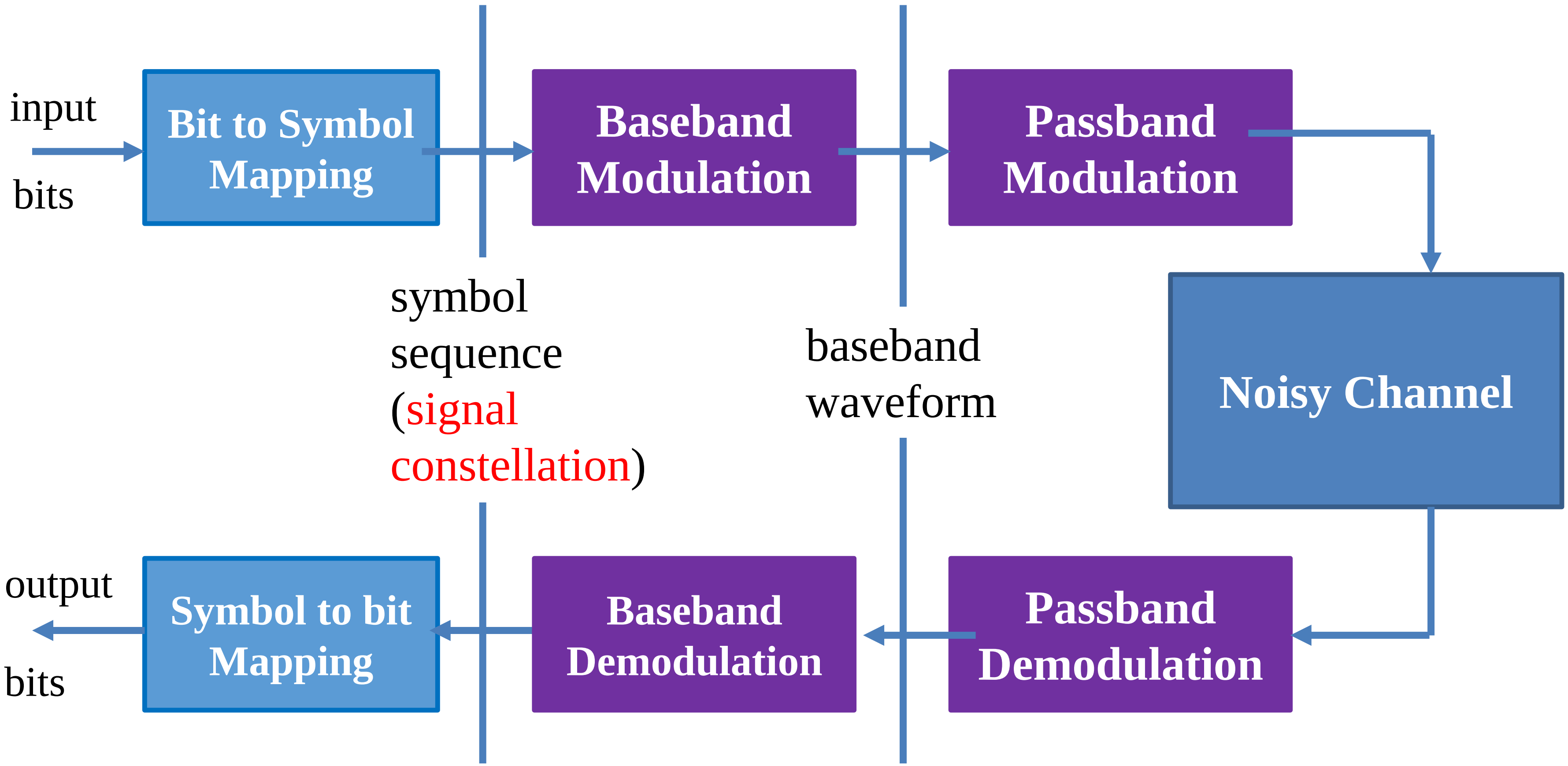
A linear operation on a_i



$$n \sim \mathcal{N}(0, \sigma^2)$$

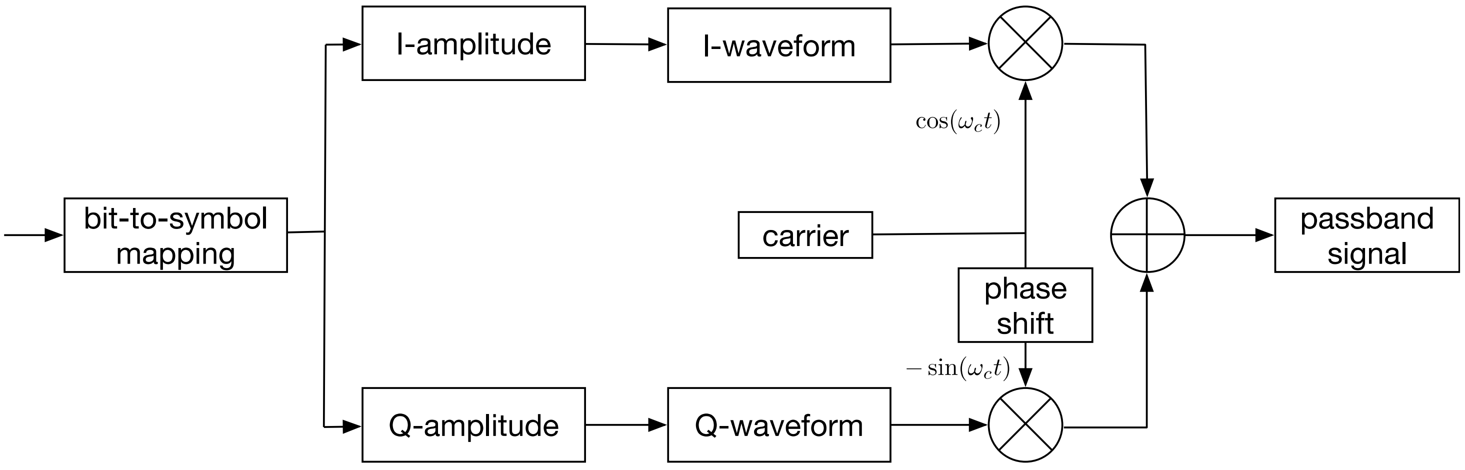
$$\sigma^2 = \frac{n_0}{2\int_{-\infty}^{\infty} |G(f)|^2 df}$$

Modulation: Overview



Passband Modulator

- General I/Q Modulator



$$x(t) = \left[\sum_n a_n g(t - nT_s)\right] \cos \omega_c t + \left[\sum_n b_n g(t - nT_s)\right] (-\sin \omega_c t)$$

$b_n = 0$: **ASK**

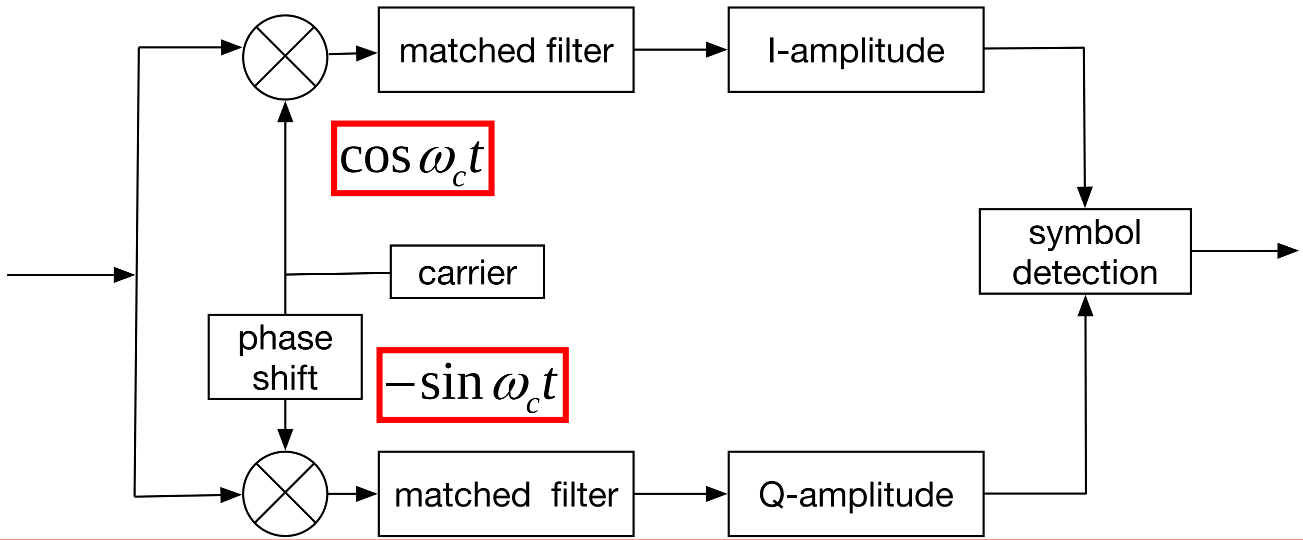
$a_n = A\cos\phi_n$, $b_n = A\sin\phi_n$: **PSK**

$$a_n \in \{\pm A, \pm 3A, \cdots, \pm(\sqrt{M} - 1)A\}$$
$$b_n \in \{\pm A, \pm 3A, \cdots, \pm(\sqrt{M} - 1)A\}$$

QAM

Passband Demodulator

- Coherent Demodulator



$$S(t)\cos\omega_c t = \left[\sum_n a_n g(t-nT_s)\right]\cos\omega_c t\cos\omega_c t + \left[\sum_n b_n g(t-nT_s)\right]\left(-\sin\omega_c t\right)\cos\omega_c t$$

$$\cos\omega_c t\cos\omega_c t = \frac{1}{2}\left[1+\cos 2\omega_c t\right]$$

$$\sin\omega_c t\cos\omega_c t = \frac{1}{2}\sin 2\omega_c t$$

Removed by LPF/matched Filter

Baseband Representation of the Passband Signal

- Goal: reuse our knowledge from baseband transmissions
- Baseband-Equivalent Complex Signal:
  - $x_{bb}(t) \triangleq x_I(t) + jx_Q(t)$
- Analytic-Equivalent Signal
  - $x_A(t) \triangleq x_{bb}(t)e^{j\omega_c t}$
  - $x(t) = \Re\{x_A(t)\} = \Re\{x_{bb}(t)e^{j\omega_c t}\}$

<u>Original passband signal is the real part of the equivalent *rotating* complex baseband representation.</u>

1. Determine passband signal from baseband signal.

Passband AWGN Channel

- Ready for the analysis
- For 1D modulation (M-ASK, M-FSK)

$$c = \int_{-\infty}^{+\infty} g^2(t)dt$$

  - $E_s = c\mathbb{E}(|\boldsymbol{a}|^2)$
  - $\sigma^2 = \frac{n_0}{2}$

$$\boxed{\frac{E_s}{n_0} = \frac{c\overline{a_n^2}}{2\sigma^2}}$$

Same as baseband

$$\boxed{\frac{c\overline{a_n^2}}{\sigma^2} = \frac{S}{N} = \frac{2E_s}{n_0}}$$

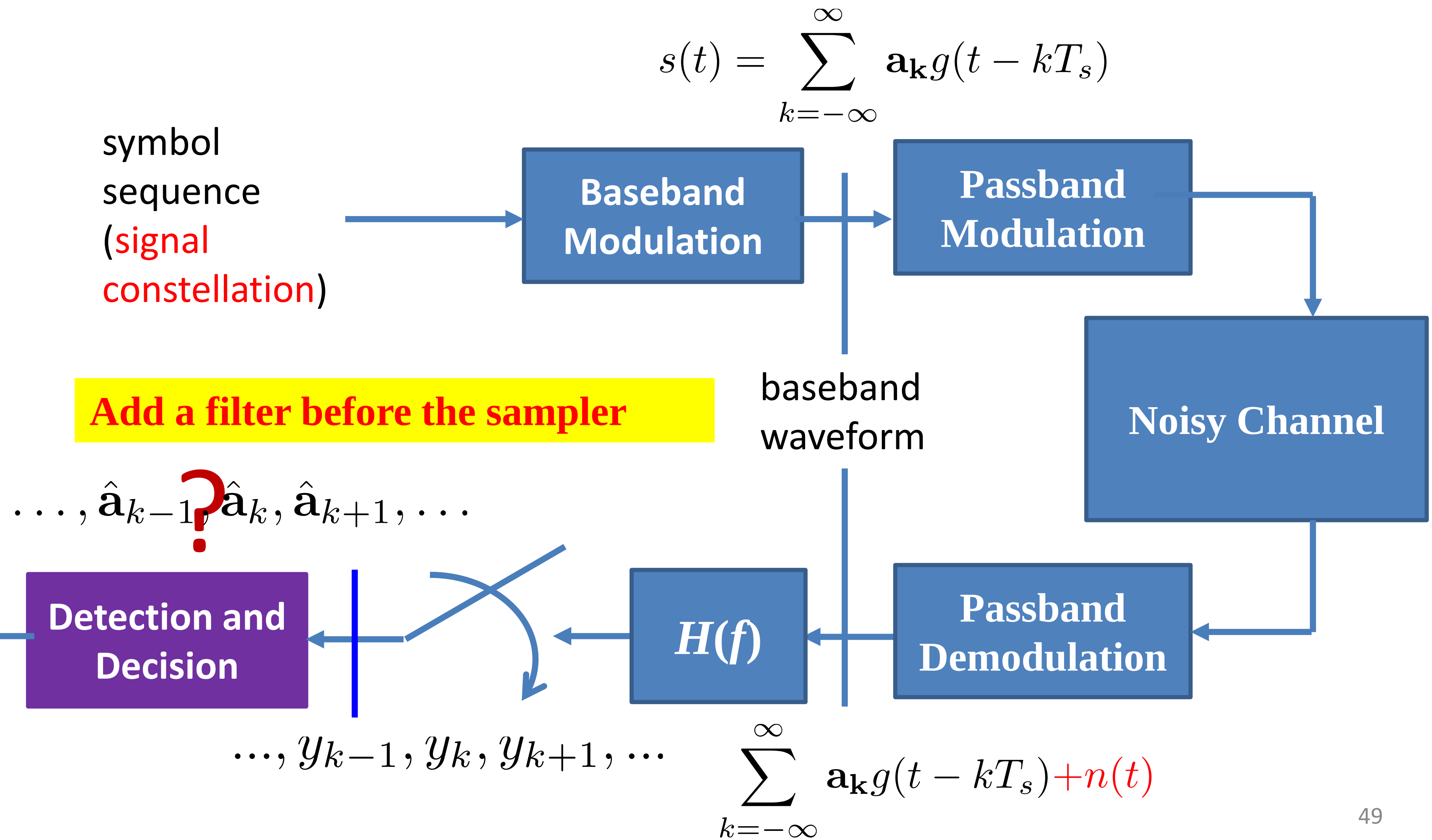
- For 2D modulation (M-PSK, M-QAM): Different!

$$\boxed{y_n = \sqrt{c}a_n + j\sqrt{c}b_n + n_I + jn_Q} \qquad \mathbf{a} = a_n + jb_n$$

  - $E_s = c\mathbb{E}(|\boldsymbol{a}|^2) = c\mathbb{E}(|a_n|^2 + |b_n|^2)$
  - $\sigma_I^2 = \sigma_Q^2 = \frac{n_0}{2}$

$$\boxed{\frac{E_s}{n_0/2} = \frac{c\overline{(a_n^2 + b_n^2)}}{\sigma_I^2} = \frac{c\overline{(a_n^2 + b_n^2)}}{\sigma_Q^2}}$$

Detection and Decision



MAP Detection

- **Maximize the a posteriori probability**

$$\boxed{P(\mathbf{a}|y) = \frac{f(y|\mathbf{a})P(\mathbf{a})}{f(y)}}$$

- We can drop $f(y)$, leading to:

$$\boxed{f(y\,|\,\mathbf{a}=\mathbf{A})P(\mathbf{a}=\mathbf{A}) \mathop{\gtrless}_{\mathbf{B}}^{\mathbf{A}} f(y\,|\,\mathbf{a}=\mathbf{B})P(\mathbf{a}=\mathbf{B})}$$

$$\boxed{\frac{f(y\,|\,\mathbf{a}=\mathbf{A})}{f(y\,|\,\mathbf{a}=\mathbf{B})} \mathop{\gtrless}_{\mathbf{B}}^{\mathbf{A}} \frac{P(\mathbf{a}=\mathbf{B})}{P(\mathbf{a}=\mathbf{A})}}$$

Only need to compare the conditional probability here

Maximum Likelihood (ML) Detection

- If we assume equal probability of all symbols (this assumption generally holds throughout this course)
  - ► For the binary case

$$P(\mathbf{a} = \mathbf{A}) = P(\mathbf{a} = \mathbf{B})$$

  - ► Then we can transform the MAP rule to

$$\boxed{\frac{f(y \mid \mathbf{a} = \mathbf{A})}{f(y \mid \mathbf{a} = \mathbf{B})} \underset{\mathbf{B}}{\overset{\mathbf{A}}{\gtrless}} 1}$$

***y* looks like to whom among –A and A most?**

  - ► This is called the Maximum Likelihood (ML) rule
  - ► Only when the probability of the hypothesis are the same, MAP is equivalent to ML

ML for the Additive Gaussian Channel

- The conditional probability is

$$\boxed{f(y\,|\,\mathbf{a}=\mathbf{A}) = \frac{1}{2\pi\sigma^2}\exp\left(-\frac{(y-\mathbf{A})^2}{2\sigma^2}\right)}$$

$$f(y\,|\,\mathbf{a}=\mathbf{B}) = \frac{1}{2\pi\sigma^2}\exp\left(-\frac{(y-\mathbf{B})^2}{2\sigma^2}\right)$$

- **With additive Gaussian Noise, we have the *minimum distance* detection rule**

$$\boxed{\frac{f(y\,|\,\mathbf{a}=\mathbf{A})}{f(y\,|\,\mathbf{a}=\mathbf{B})} \underset{\mathbf{B}}{\overset{\mathbf{A}}{\gtrless}} 1} \Rightarrow \boxed{|\,y-\mathbf{A}\,| \underset{\mathbf{A}}{\overset{\mathbf{B}}{\gtrless}} |\,y-\mathbf{B}\,|}$$

Review the Steps of Calculating SEP

- 1. Draw the symbols
  - With inter-symbol distance $2A$
- 2. Decide the detection thresholds
  - Distance between the symbol to its thresholds is A
- 3. Derive the conditional error probability for each symbol as the function of A/σ
  - Case by Case
- 4. <u>Use A to characterize E_s</u>
  - Obtain A/σ as the function of $\boldsymbol{E_s/n_0}$

Choose correct σ !! (from which dimension?)

- 5. Get the SEP as a function of E_s/n_0

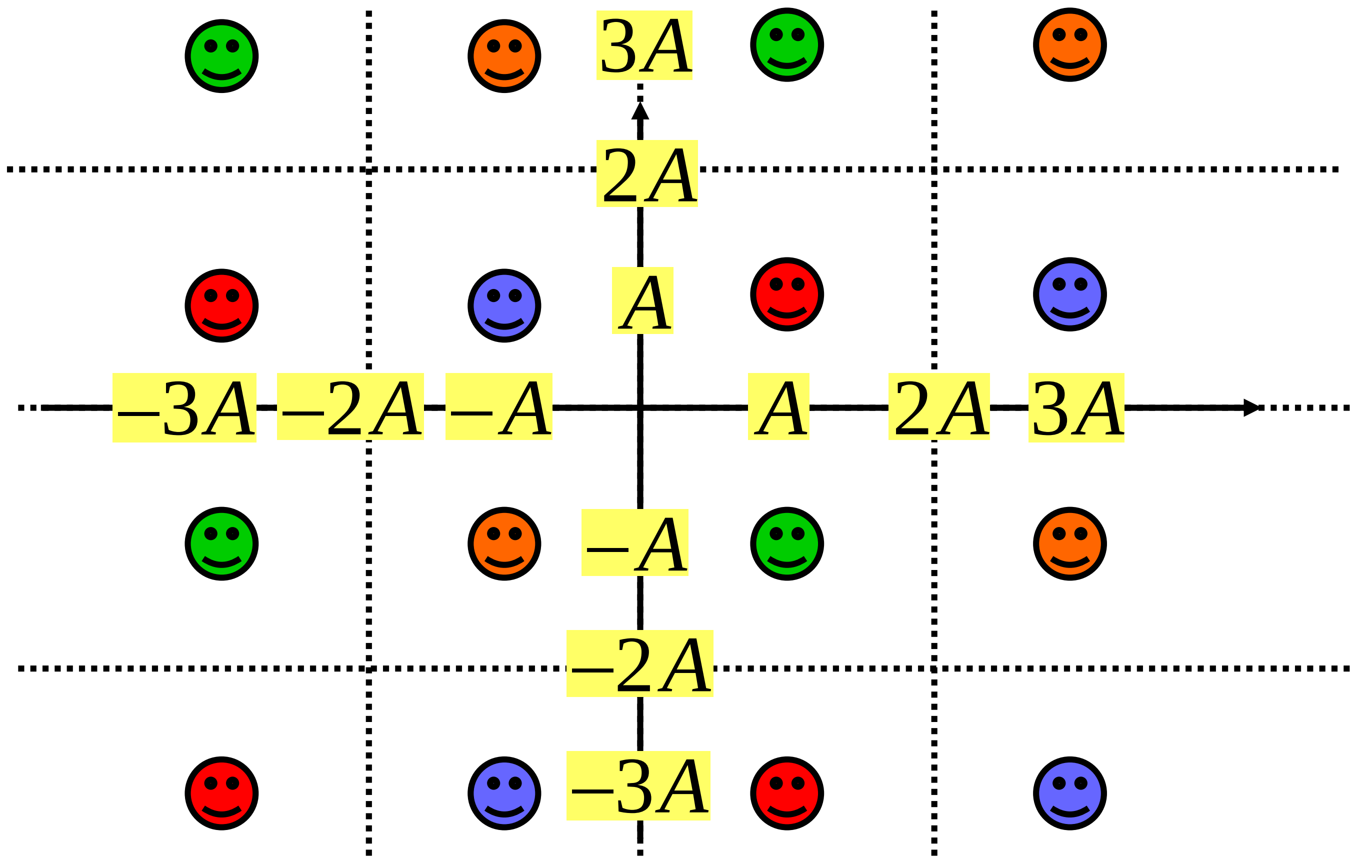
1-D $\quad \dfrac{E_s}{n_0} = \dfrac{c\overline{a_n^2}}{2\sigma^2}$

2-D $\quad \dfrac{E_s}{n_0} = \dfrac{c\overline{(a_n^2 + b_n^2)}}{\sigma^2}$

$\dfrac{E_s}{n_0/2} = \dfrac{c\overline{(a_n^2 + b_n^2)}}{\sigma_I^2}$

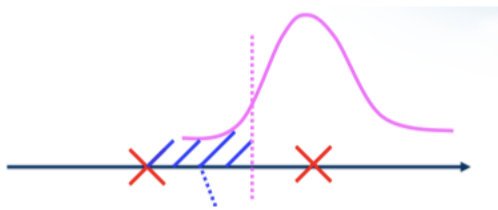
MQAM Error Analysis

- For MQAM, the detection threshold is a grid.



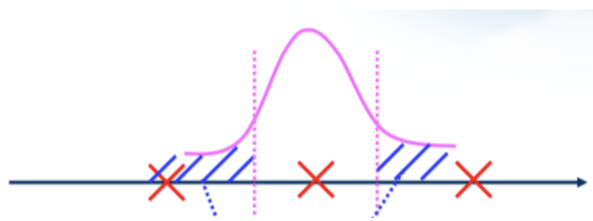
MQAM Error Analysis

- Recall our building block:
  - Uni-lateral Error



$$Q\left(\frac{A}{\sigma_L}\right)$$

  - Bilateral Error



$$2Q\left(\frac{A}{\sigma_L}\right)$$

- Consider MQAM as two linear ASK

$$\sigma_L = \sigma_I = \sigma_Q = \sqrt{\frac{n_0}{2}} = \frac{\sigma}{\sqrt{2}}$$

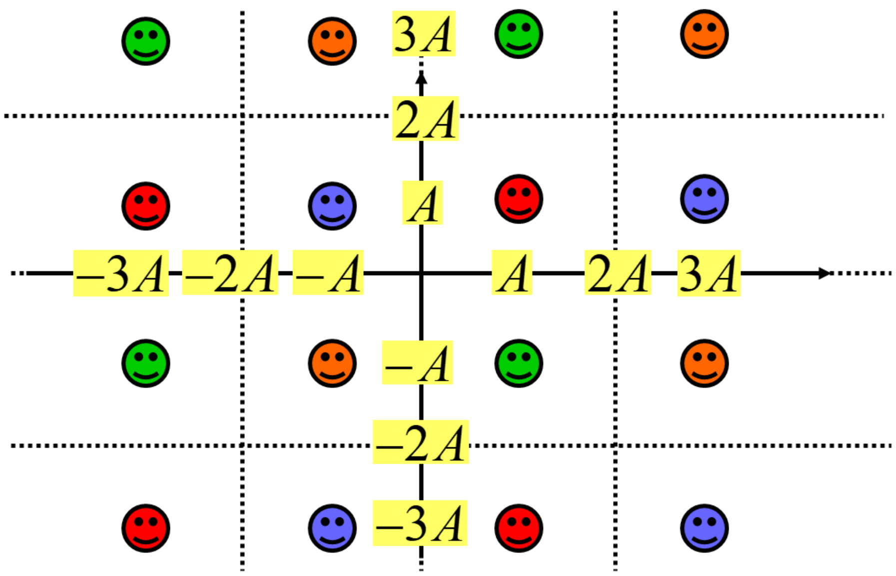
MQAM Error Analysis

- 4 points at the corner
  - For each dimension, the probability of no error is

$$1 - Q\left(\frac{A}{\sigma_L}\right)$$

  - The symbol error probability is

$$P_s = 1 - \left[1 - Q\left(\frac{A}{\sigma_L}\right)\right]^2$$



MQAM Error Analysis

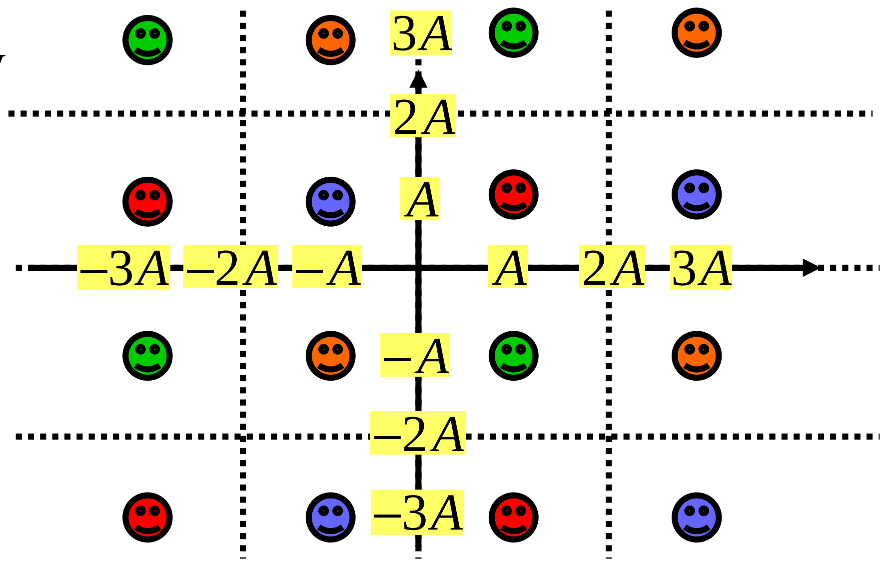
- $4(L-2)$ points on 4 sides, $L=\sqrt{M}$
  - For one dimension, the probability of no error is

$$1-Q\left(\frac{A}{\sigma_L}\right)$$

  - For another dimension,

$$1-2Q\left(\frac{A}{\sigma_L}\right)$$

  - Hence



$$P_s = 1-\left[1-Q\left(\frac{A}{\sigma_L}\right)\right]\left[1-2Q\left(\frac{A}{\sigma_L}\right)\right]$$

MQAM Error Analysis

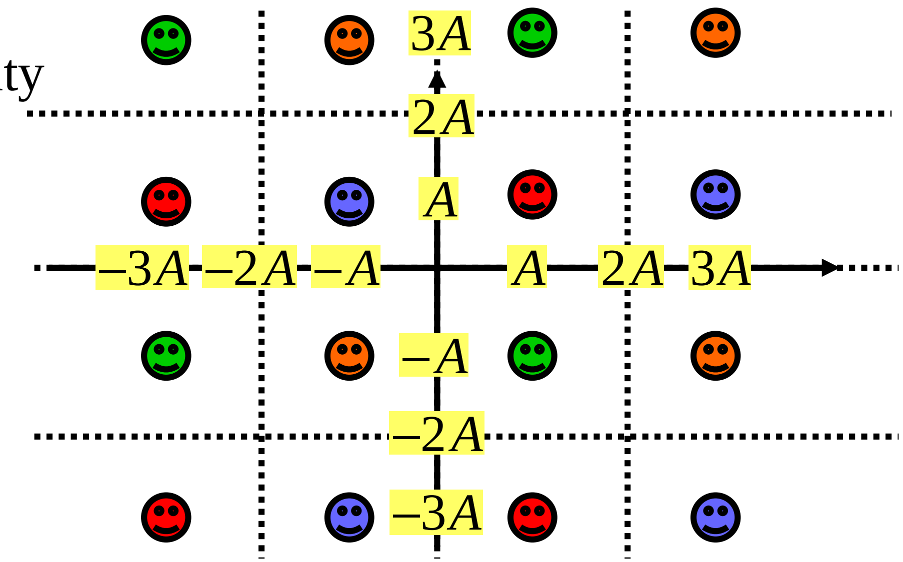
- $(L-2)^2$ internal points

  - For each dimension, the probability of no error is

$$1 - 2Q\left(\frac{A}{\sigma_L}\right)$$

  - The symbol error rate is

$$P_s = 1 - \left[1 - 2Q\left(\frac{A}{\sigma_L}\right)\right]^2$$



MQAM Error Analysis

- Average SEP

$$P_s = \frac{1}{L^2}\left[4(2Q - Q^2) + 4(L-2)(3Q - 2Q^2) + (L-2)^2(4Q - 4Q^2)\right]$$
$$\approx \frac{1}{L^2}\left[8Q + 12(L-2)Q + 4(L-2)^2 Q\right]$$
$$= \frac{4L^2 - 4L}{L^2} Q\left(\frac{A}{\sigma_L}\right)$$

- Proakis Method

$$P_s = 1 - \left(1 - P_{s,LPAM}\right)^2 = \frac{4L^2 - 4L}{L^2} Q\left(\frac{A}{\sigma_L}\right)$$

MQAM Error Analysis

- Represent E_s with A

$$\begin{aligned} E_s &= 2 \times \frac{2A^2}{L}[1^2 + 3^2 + \cdots + (L-1)^2] \\ &= 2 \times \frac{2A^2}{L}\frac{L}{6}(L^2 - 1) \\ &= \frac{2(L^2 - 1)}{3}A^2 \end{aligned}$$

MQAM Error Analysis

$$\boxed{\sigma_L = \sigma_I = \sigma_Q = \sqrt{\frac{n_0}{2}}}$$

- Express SEP with E_s/n_0

$$\frac{A}{\sigma_L} = \sqrt{\frac{3}{L^2 - 1}\frac{E_s}{n_0}}$$

$$P_{s,MQAM} = \frac{4L^2 - 4L}{L^2} Q\left(\sqrt{\frac{3}{L^2 - 1}\frac{E_s}{n_0}}\right)$$

MQAM Error Analysis

- Express SEP with E_b/N_0

$$P_{s,MQAM} = 4\left(1 - \frac{1}{\sqrt{M}}\right) Q\left(\sqrt{\frac{3\log_2 M}{M-1}\frac{E_b}{n_0}}\right) \approx 4Q\left(\sqrt{\frac{3\log_2 M}{M-1}\frac{E_b}{n_0}}\right)$$

Large M approximation

- Convert from SEP to BEP

$$P_{b,MQAM} = \frac{4}{\log_2 M} Q\left(\sqrt{\frac{3\log_2 M}{M-1}\frac{E_b}{n_0}}\right)$$

Thank you !

(Q&A)